

Sprint Report

Portfolio Task 2/3/4

Unit code: COS40005/EAT40005

Unit name: Computing/Engineering Technology Project A

Submission date: 7th of November, 2025

Group 7:

Daniel Tiong (102777801)

Alif Harriz (102782711)

Ashley Jong (102780087)

Basil Agas (102778888)

Edison Ho (102779496)

1. Contribution Summary (This Sprint)

All team members should complete the following table together. You can provide a yes/no response or a brief comment where appropriate.

Team Member	Daniel Tiong	Alif Harriz	Ashley Jong	Basil Agas	Edison Ho
Contribution Finished tasks in time or on time with an acceptable quality	Yes	Yes	Yes	Yes	Yes
Initiatives Self-assigned tasks and proactively found solutions to problems	Yes	Yes	Yes	Yes	Yes
Communication Attended all team meetings	Yes	Yes	Yes	Yes	Yes
Communication Attended all supervisor meetings	Yes	Yes	Yes	Yes	Yes
Communication	Yes	Yes	Yes	Yes	Yes

Attended all client meetings					
Communication					
Replied every time within an acceptable timeframe	Yes	Yes	Yes	Yes	Yes
Communication					
Kept the team updated about the status	Yes	Yes	Yes	Yes	Yes
Respect					
The student respected others and listened to different opinions	Yes	Yes	Yes	Yes	Yes

Table 1: Contribution summary for Sprint 2

2. Contribution Details (This Sprint)

In this section, every member of the team should write a summary of what they have contributed during this sprint. All contributions must be supported by evidence, such as the number of GitHub commits, a screenshot of developed screens, a summary/outcome of research, etc.

2.1. Daniel (Project Lead and AI Engineer)

As Project Lead, I orchestrated the sprint's main objective: the critical architectural refactoring of the project into a monorepo. I managed the GitHub repository, which included merging the pull request for the clinician dashboard and ensuring a smooth transition to the new structure.

My technical work involved both backend integration and AI development. I collaborated closely with Edison and the frontend team to connect the different services, with a focus on testing the user authentication flow end-to-end. To support this, I developed a Python-based devtool for comprehensive local and online API testing. I deployed our backend services to Vercel and conducted tests to ensure the connections were stable. Additionally, I did preliminary work on the LLM action proof-of-concept, confirming the viability of our proposed AI architecture.

Evidence of Work

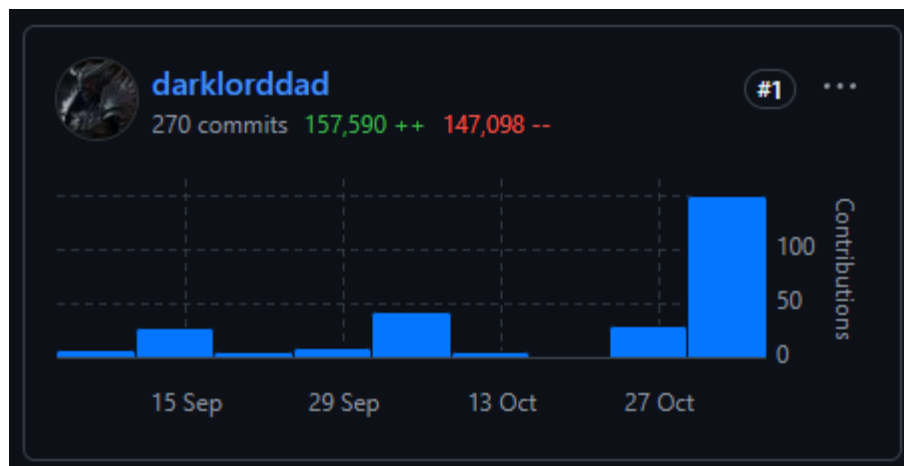


Figure 1: GitHub contribution metrics

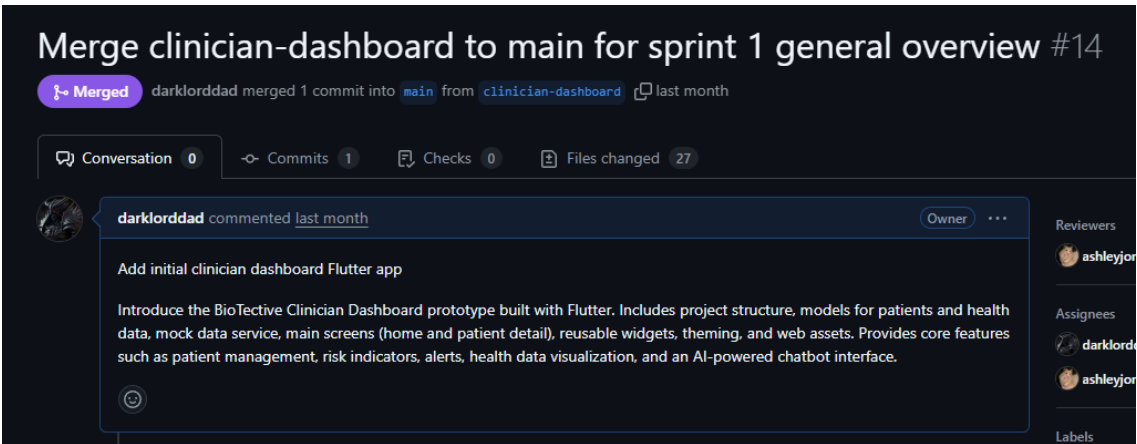


Figure 2: Pull request merged

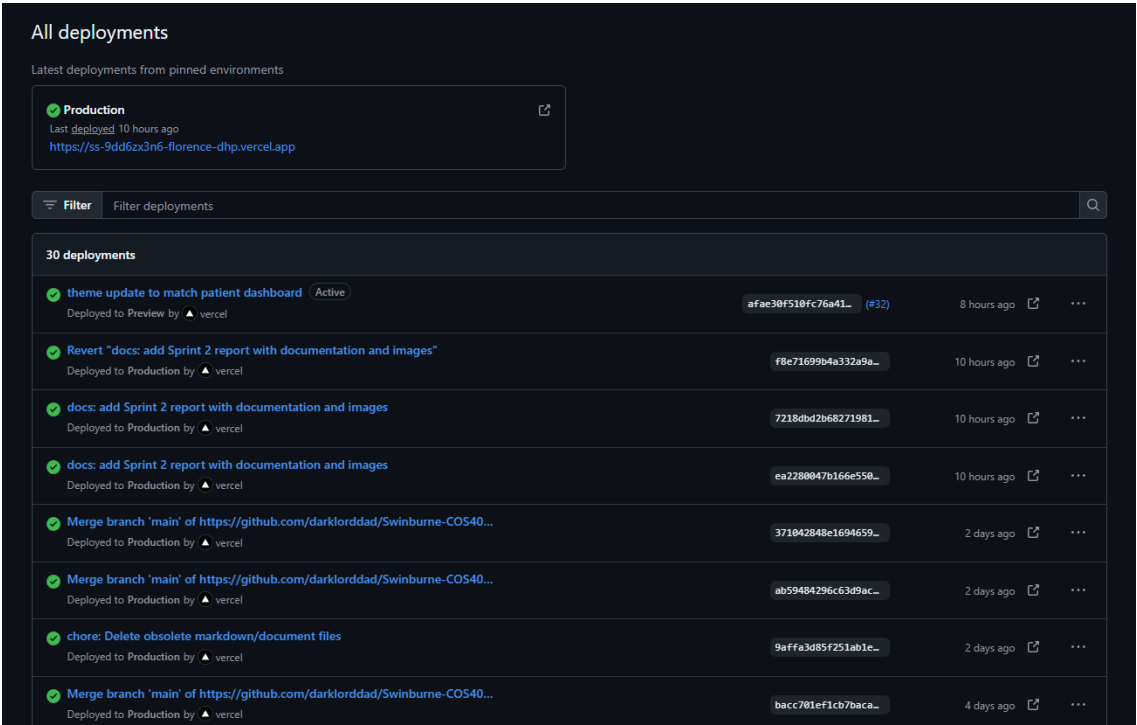


Figure 3: GitHub to Vercel deployment

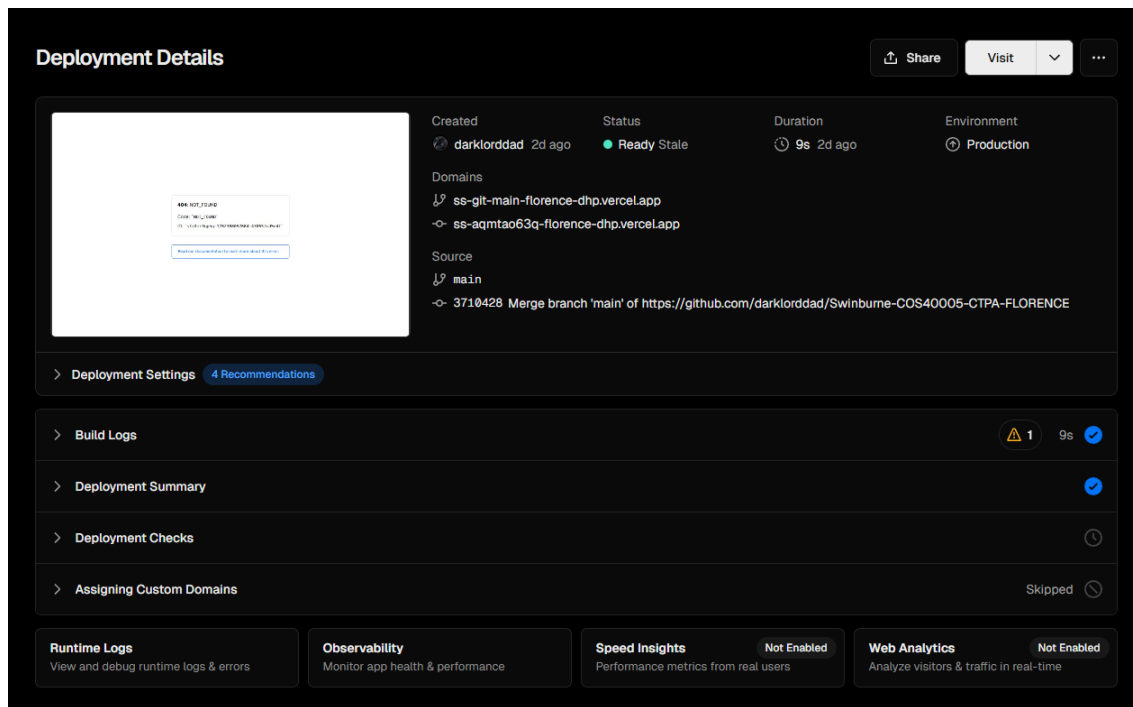


Figure 4: Vercel deployment

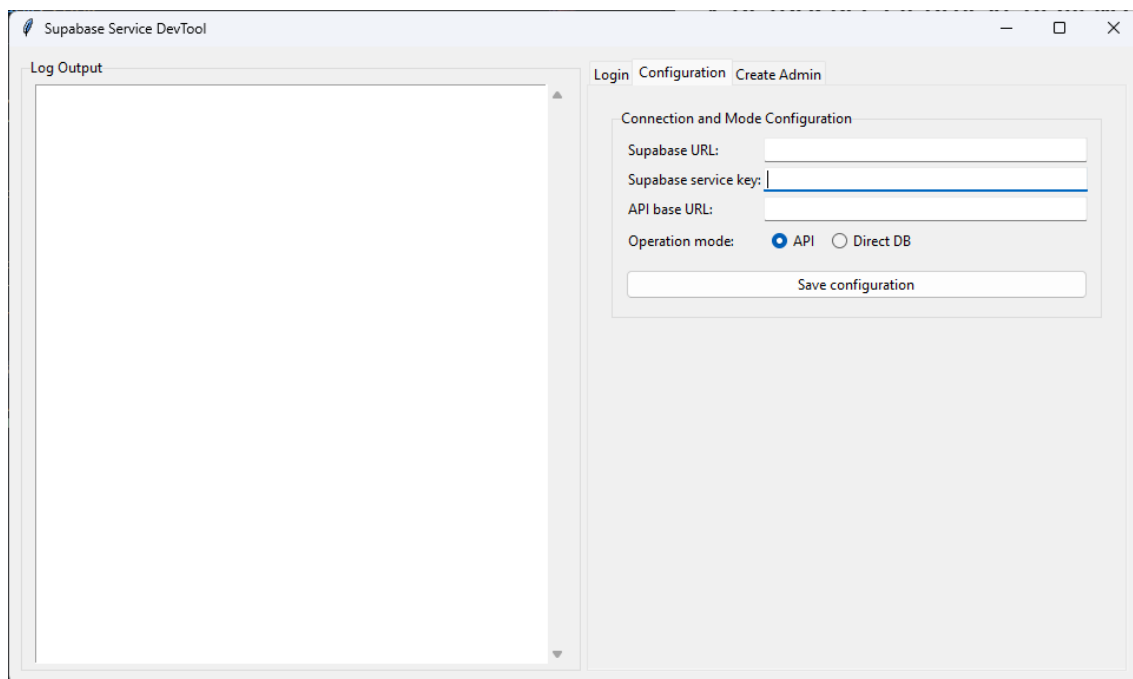


Figure 5: Supabase DevTool for testing purposes

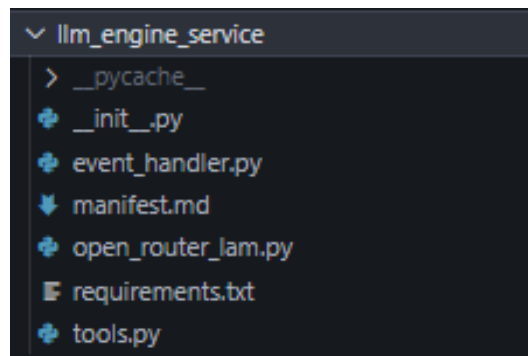


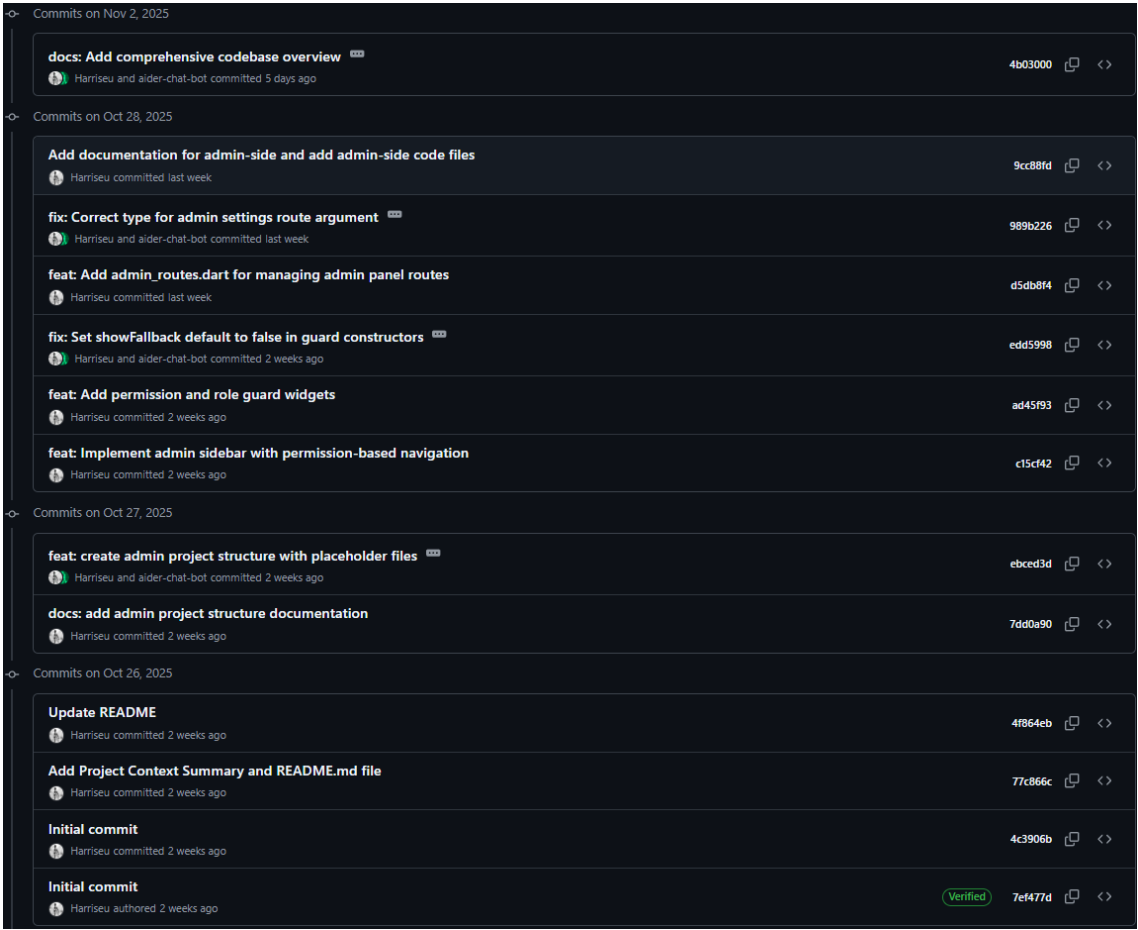
Figure 6: AI component

2.2. Alif (Frontend Developer - Patient Dashboard)

My primary focus this sprint was leading the development of the patient-facing dashboard UI in Flutter. I built the foundational application prototype, including the main navigation structure and a placeholder for the future chatbot integration. A key part of my work involved integrating the *fl_chart* library with dummy data to create the responsive data visualisations for glucose trends, diet, and activity, ensuring the layout works effectively on both mobile and desktop views.

Evidence of Work

GitHub commits on my repository in collaboration with Basil for AI Implementation



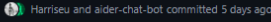
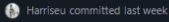
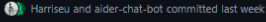
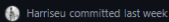
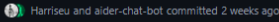
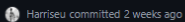
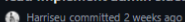
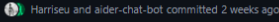
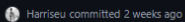
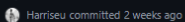
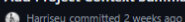
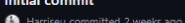
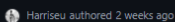
Commits on Nov 2, 2025	
docs: Add comprehensive codebase overview 	4b03000  
	
Commits on Oct 28, 2025	
Add documentation for admin-side and add admin-side code files	9cc88fd  
	
fix: Correct type for admin settings route argument 	989b226  
	
feat: Add admin_routes.dart for managing admin panel routes	d5db8f4  
	
fix: Set showFallback default to false in guard constructors 	edd5998  
	
feat: Add permission and role guard widgets	ad45f93  
	
feat: Implement admin sidebar with permission-based navigation	c15cf42  
	
Commits on Oct 27, 2025	
feat: create admin project structure with placeholder files 	ebced3d  
	
docs: add admin project structure documentation	7dd0a90  
	
Commits on Oct 26, 2025	
Update README	4f864eb  
	
Add Project Context Summary and README.md file	77c866c  
	
Initial commit	4c3906b  
	
Initial commit	 7ef477d  
	

Figure 7: Repository commits (Patient-side and AI implementation)

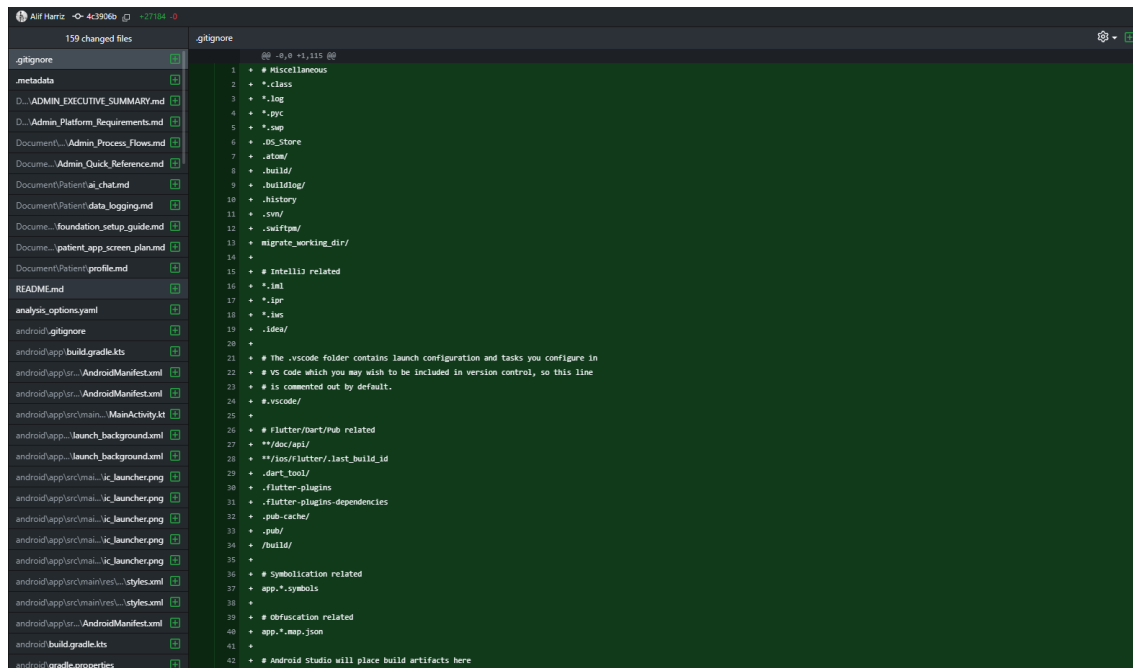


Figure 8: Patient-side and AI implementation commit history

GitHub commits on the main repository (transferred all codes from my repository to the main repository):

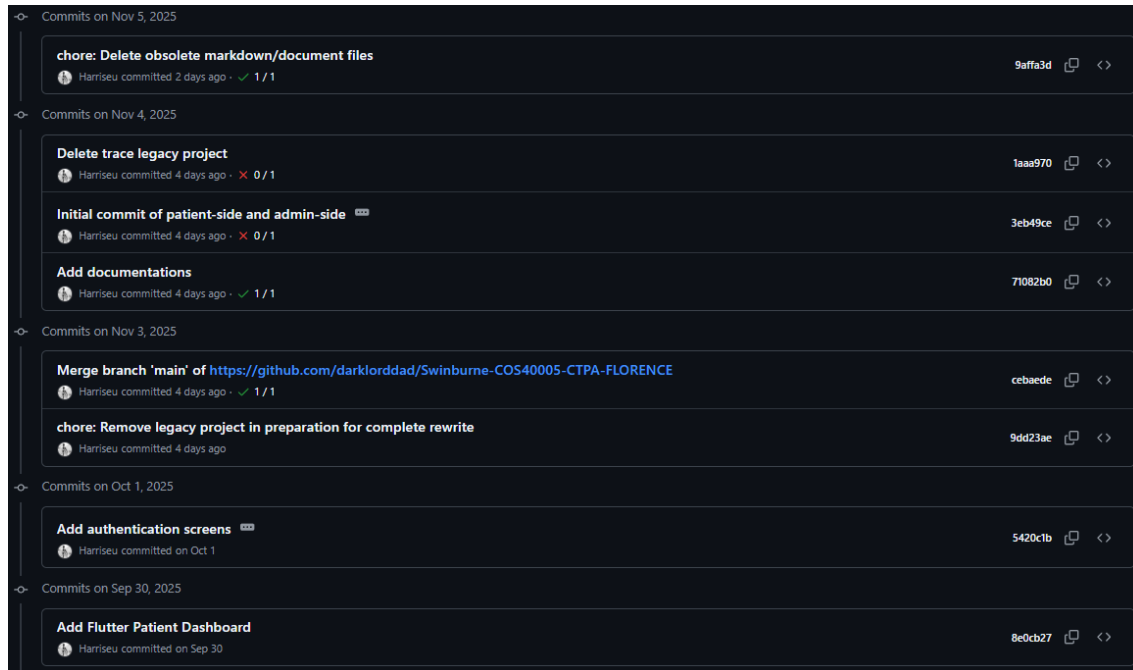


Figure 9: Main repository commits

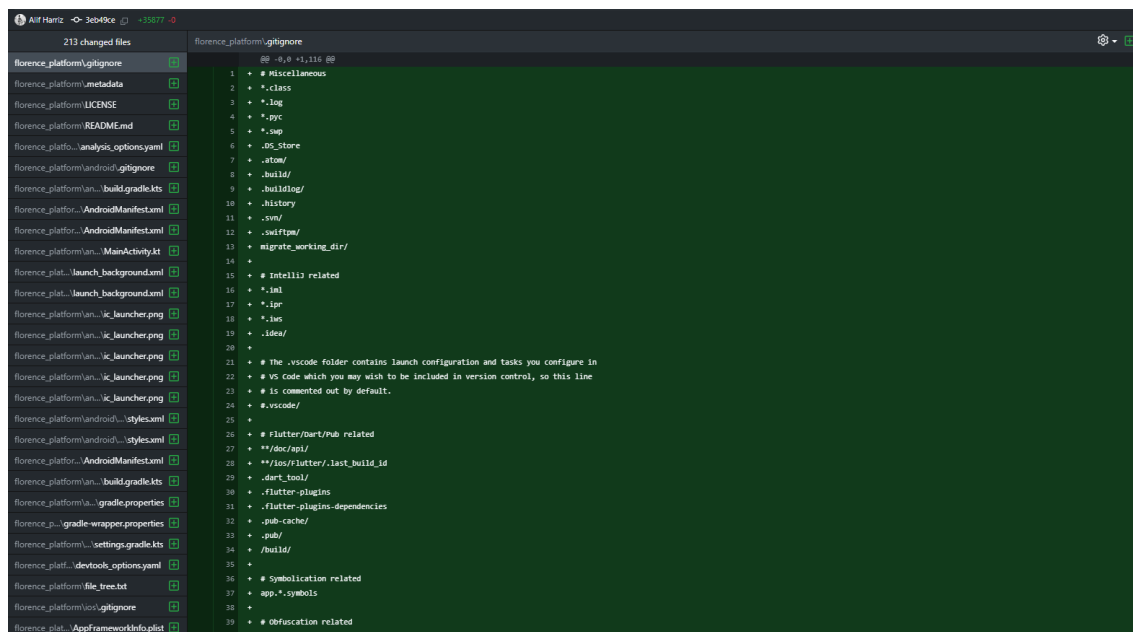
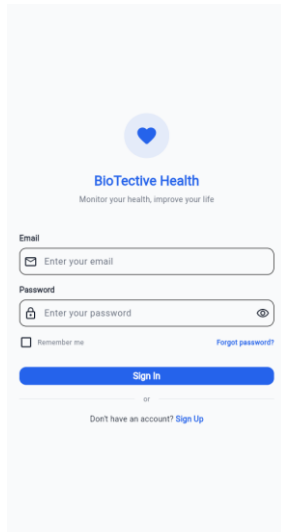


Figure 10: Main repository commit history

Screenshots of patient-side application:



BioTective Health
Monitor your health, improve your life

Email
Enter your email

Password
Enter your password

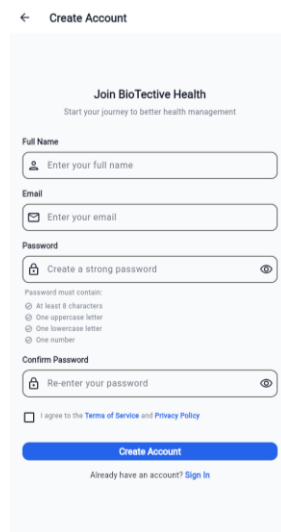
☐ Remember me [Forgot password?](#)

[Sign In](#)

or

Don't have an account? [Sign Up](#)

Figure 11: Login and sign-up screen



Create Account

Join BioTective Health
Start your journey to better health management

Full Name
Enter your full name

Email
Enter your email

Password
Create a strong password

Password must contain:
☐ At least 8 characters
☐ One uppercase letter
☐ One lowercase letter
☐ One number

Confirm Password
Re-enter your password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

[Create Account](#)

Already have an account? [Sign In](#)

Figure 12: Sign-up screen

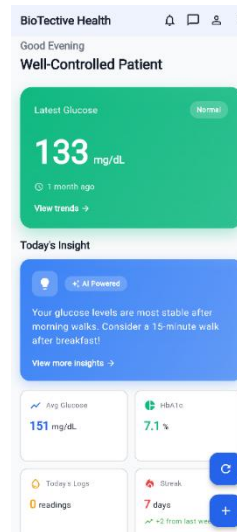


Figure 13: Patient main dashboard

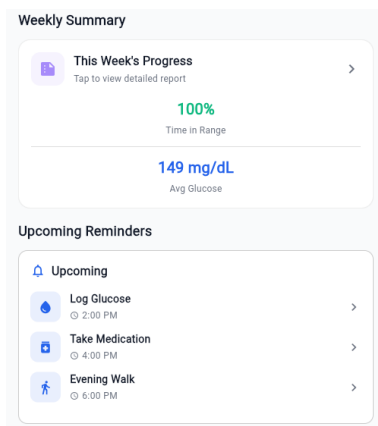


Figure 14: Weekly summary and upcoming reminders screen

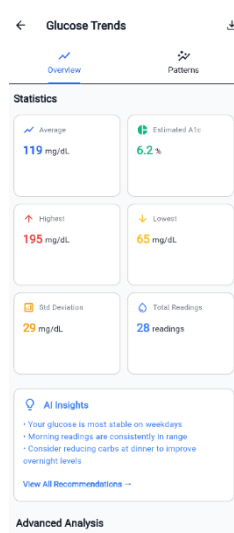
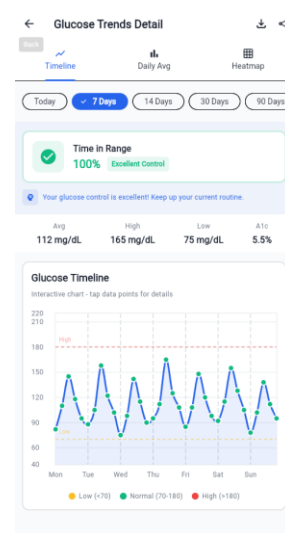
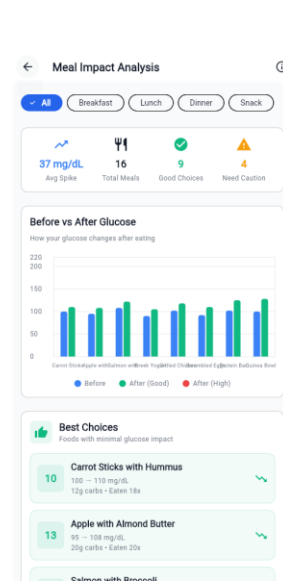
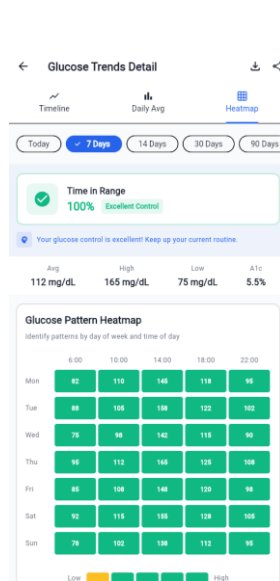
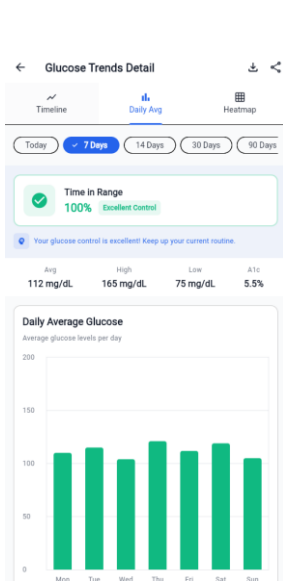


Figure 15: Glucose trends screen





Glucose

Glucose Trends Detail

Figure : Glucose Trend Detail Timeline

Figure : Glucose Trends Detail Daily Avg

Figure : Glucose Trends Detail Heatmap

Meal Impact Analysis

Figure : Meal Impact Analysis Before vs After

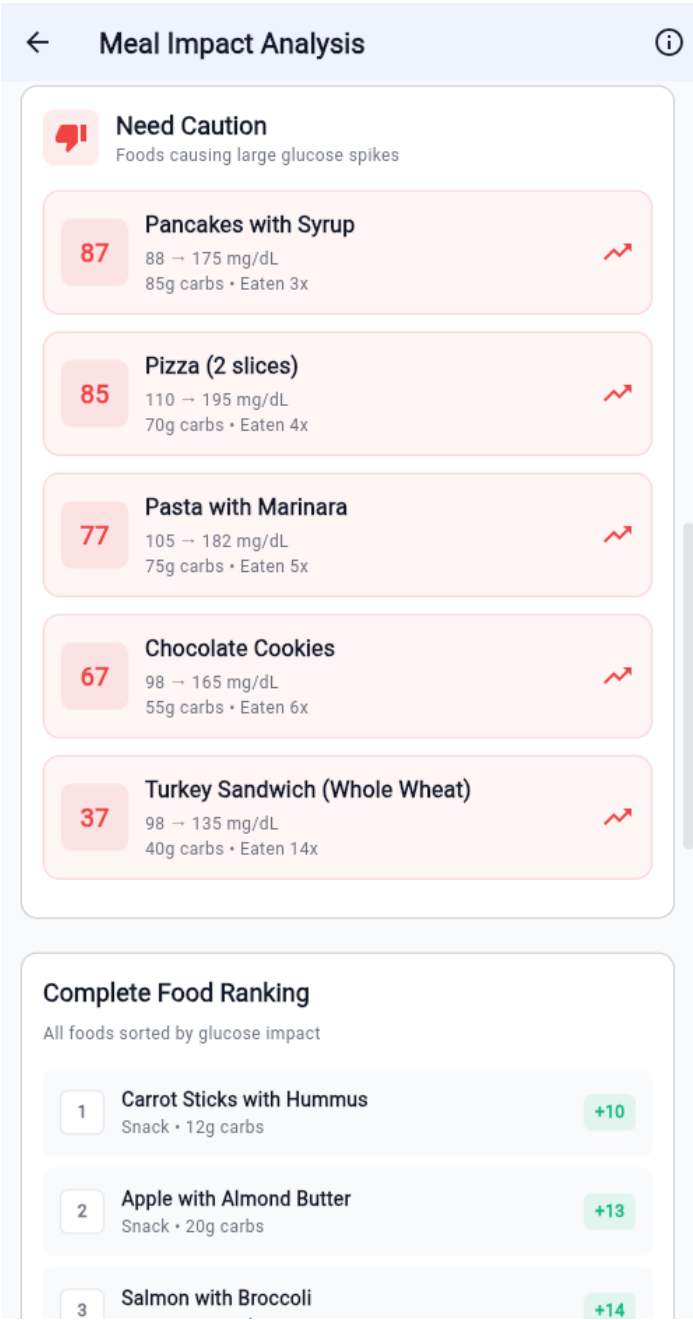


Figure : Meal Impact Analysis Need Caution

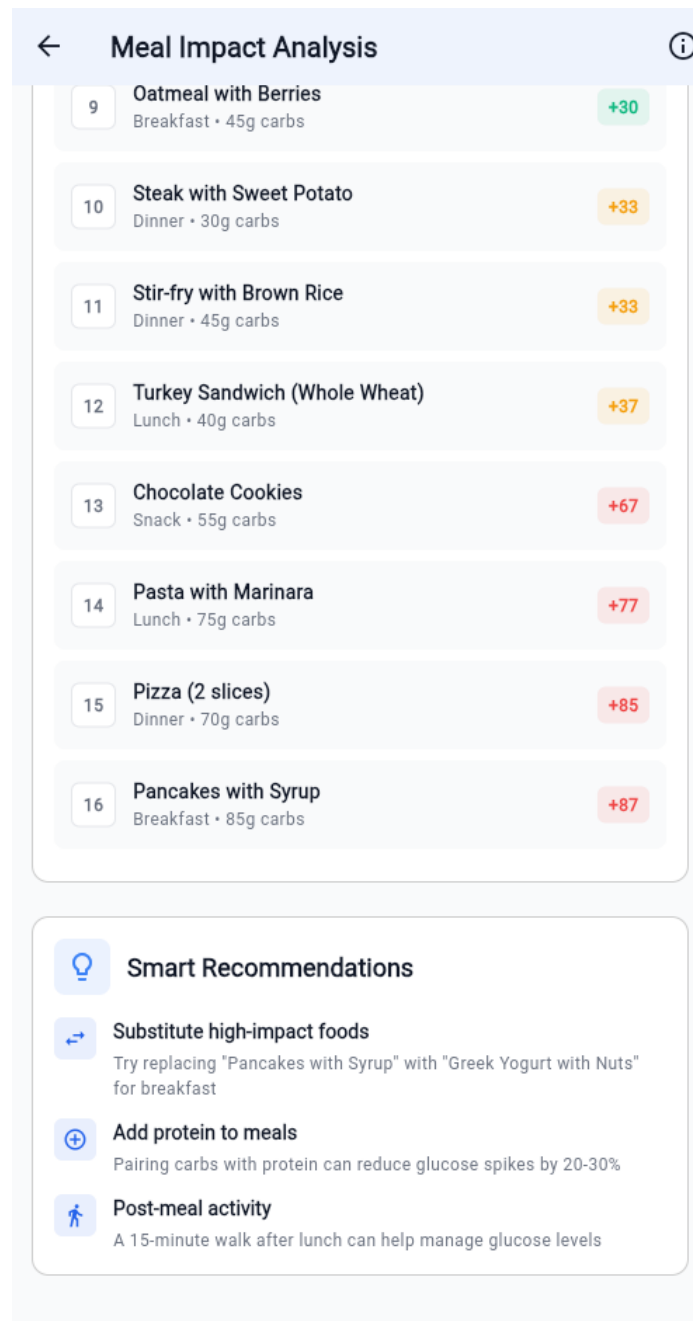


Figure : Meal Impact Analysis Recommendations

Activity Impact Analysis



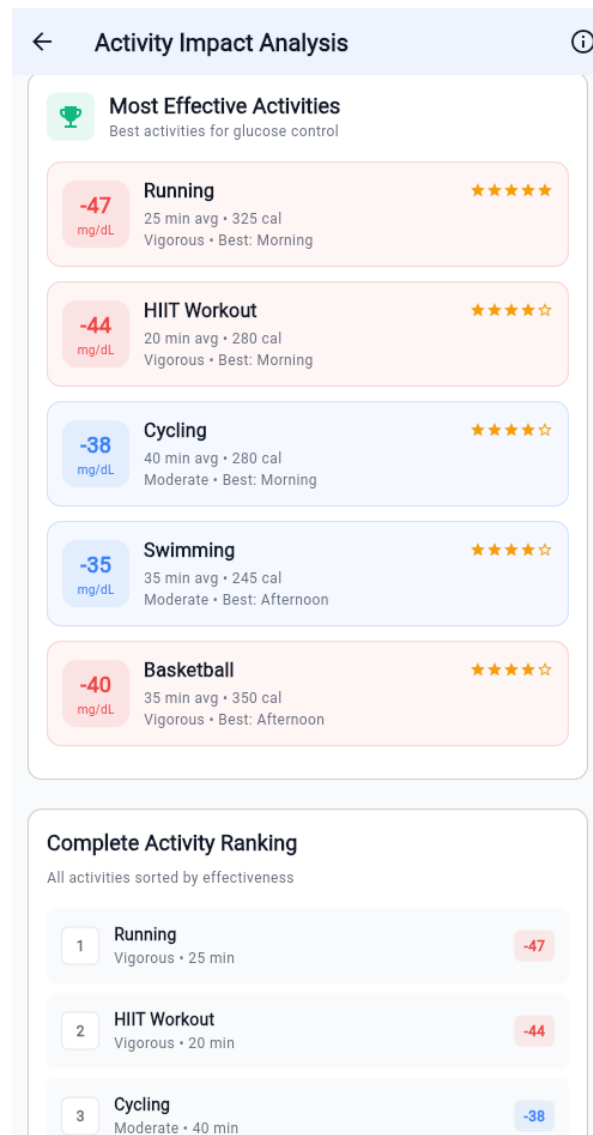


Figure : Activity Impact Analysis Effective Activities

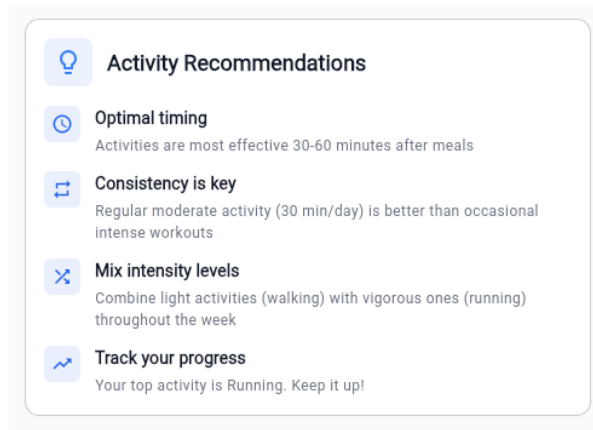


Figure : Activity Impact Analysis Effective Activities

Quick Log functionality for logging health data with ease

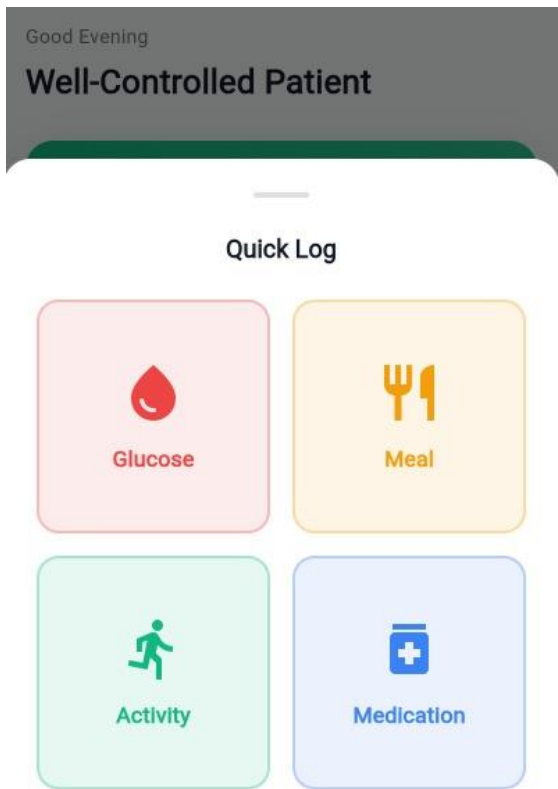


Figure : Quick Log

Patient Profile Screen

←

Profile & Settings

↗

Back

J

John Doe

nikocado@gmail.com

Edit Profile

Personal Information

Date of Birth

January 15, 1985

Gender

Male

Phone Number

+60 12-345 6789

Health Profile

Diabetes Type

Type 2

Target Glucose Range

70-180 mg/dL

Medications

+ Add

Metformin

500mg • Twice daily

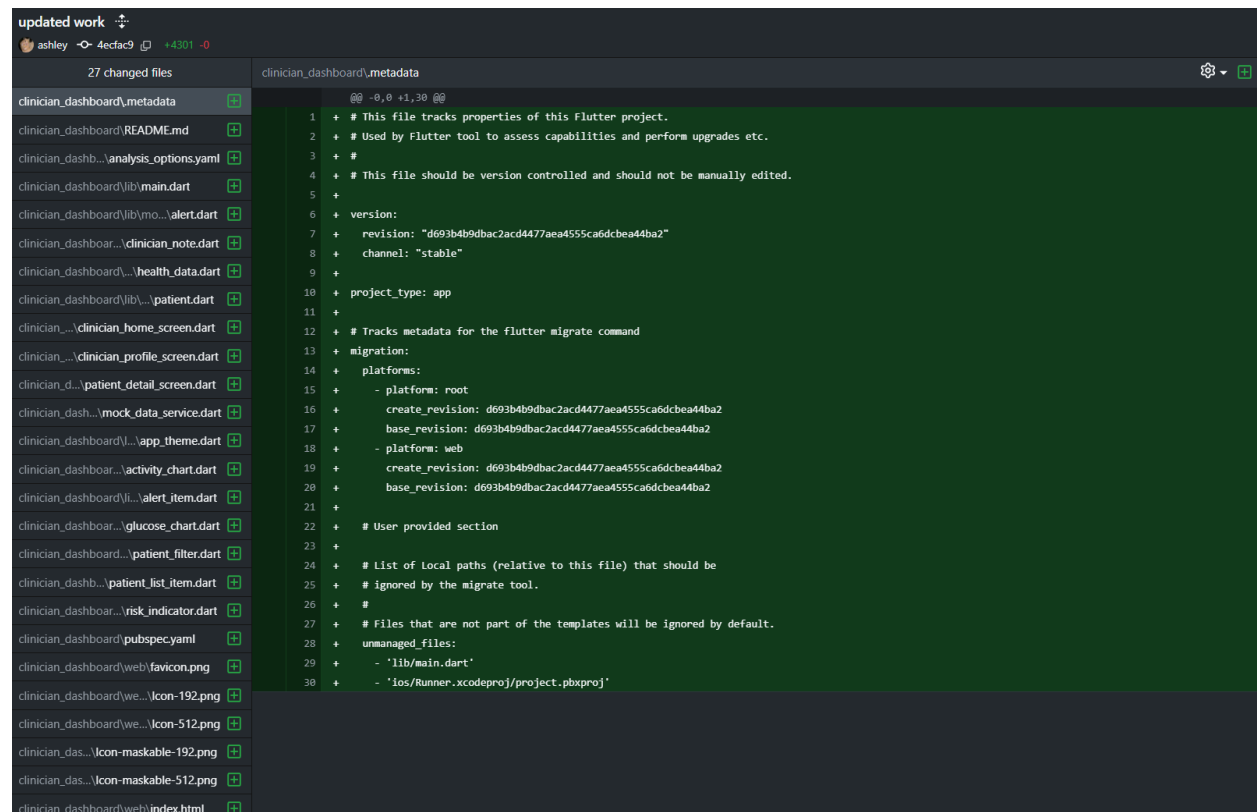
Figure : Profile Screen

2.3. Ashley (Frontend Developer - Clinician Dashboard)

This sprint, I was responsible for the entire lifecycle of the clinician-facing dashboard, from design to implementation. I began by designing and creating detailed wireframes that mapped out an intuitive user interface for viewing aggregated patient data, highlighting anomalies, and prioritising patients. Using these wireframes as a blueprint, I then coded the complete front-end of the dashboard in Flutter, translating the static designs into a responsive and interactive application.

Evidence of Work

GitHub Commits:



The screenshot displays a GitHub commit interface for a repository named 'updated work'. The commit is by user 'ashley' with commit hash '4ecfac9' and a diff of '+4301 -0'. A list of 27 changed files is shown on the left, including 'clinician_dashboard\metadata', 'clinician_dashboard\README.md', 'clinician_dashb...\analysis_options.yaml', 'clinician_dashboard\lib\main.dart', 'clinician_dashboard\lib\mo...\alert.dart', 'clinician_dashboard...\clinician_note.dart', 'clinician_dashboard...\health_data.dart', 'clinician_dashboard\lib\...\patient.dart', 'clinician...\clinician_home_screen.dart', 'clinician...\clinician_profile_screen.dart', 'clinician_d...\patient_detail_screen.dart', 'clinician_dash...\mock_data_service.dart', 'clinician_dashboard...\app_theme.dart', 'clinician_dashboard...\activity_chart.dart', 'clinician_dashboard\li...\alert_item.dart', 'clinician_dashboard...\glucose_chart.dart', 'clinician_dashboard...\patient_filter.dart', 'clinician_dashb...\patient_list_item.dart', 'clinician_dashboard...\risk_indicator.dart', 'clinician_dashboard\pubspec.yaml', 'clinician_dashboard\web\favicon.png', 'clinician_dashboard\we...\icon-192.png', 'clinician_dashboard\we...\icon-512.png', 'clinician_das...\icon-maskable-192.png', 'clinician_das...\icon-maskable-512.png', and 'clinician_dashboard\web\index.html'. The right pane shows the diff for 'clinician_dashboard\metadata', which is a Flutter project's metadata file. The diff shows the file's purpose, version control information, and migration details for both root and web platforms.

```
@@ -0,0 +1,30 @@
1 + # This file tracks properties of this Flutter project.
2 + # Used by Flutter tool to assess capabilities and perform upgrades etc.
3 + #
4 + # This file should be version controlled and should not be manually edited.
5 +
6 + version:
7 +   revision: "d693b4b9dbac2acd4477aea4555ca6dcbea44ba2"
8 +   channel: "stable"
9 +
10 + project_type: app
11 +
12 + # Tracks metadata for the flutter migrate command
13 + migration:
14 +   platforms:
15 +     - platform: root
16 +       create_revision: d693b4b9dbac2acd4477aea4555ca6dcbea44ba2
17 +       base_revision: d693b4b9dbac2acd4477aea4555ca6dcbea44ba2
18 +     - platform: web
19 +       create_revision: d693b4b9dbac2acd4477aea4555ca6dcbea44ba2
20 +       base_revision: d693b4b9dbac2acd4477aea4555ca6dcbea44ba2
21 +
22 + # User provided section
23 +
24 + # List of Local paths (relative to this file) that should be
25 + # ignored by the migrate tool.
26 + #
27 + # Files that are not part of the templates will be ignored by default.
28 + unmanaged_files:
29 +   - 'lib/main.dart'
30 +   - 'ios/Runner.xcodeproj/project.pbxproj'
```

theme update to match patient dashboard

ashley afac30f +97 -32

2 changed files

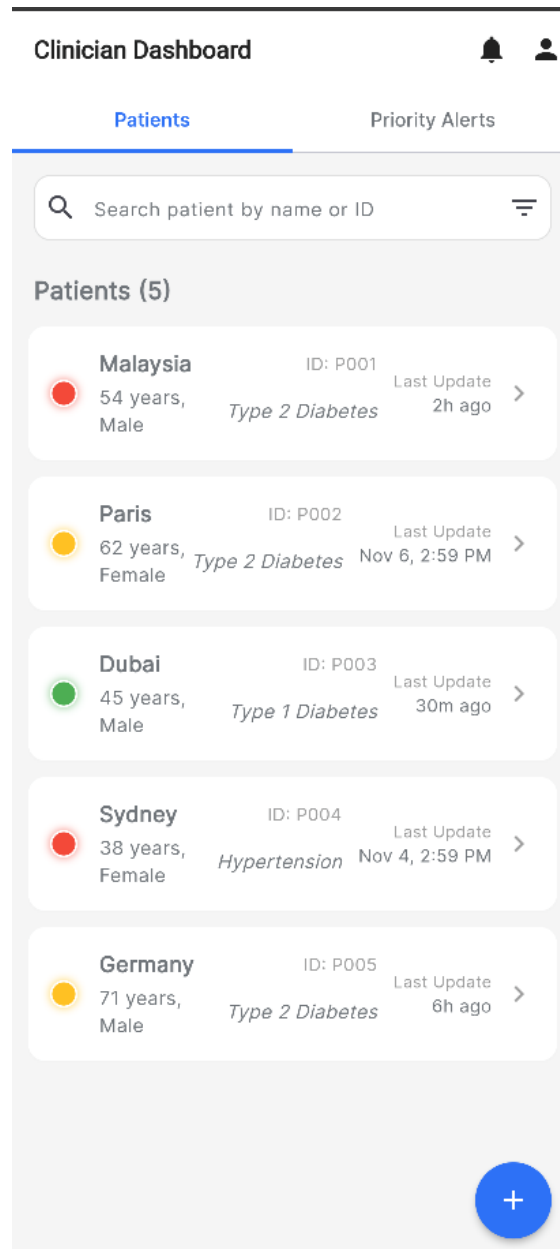
clinician_dashboard/.gitignore

clinician_dashboard/.gitignore

clinician_dashboard/.app_theme.dart

```
@@ -0,0 +1,45 @@
1 + # Miscellaneous
2 + *.class
3 + *.log
4 + *.pyc
5 + *.sap
6 + .DS_Store
7 + .atom/
8 + .build/
9 + .buildlog/
10 + .history
11 + .svn/
12 + .swiftpm/
13 + migrate_working_dir/
14 +
15 + # IntelliJ related
16 + *.iml
17 + *.ipr
18 + *.iws
19 + .idea/
20 +
21 + # The .vscode folder contains launch configuration and tasks you configure in
22 + # VS Code which you may wish to be included in version control, so this line
23 + # is commented out by default.
24 + #.vscode/
25 +
26 + # Flutter/Dart/Pub related
27 + **/doc/api/
28 + **/ios/Flutter/.last_build_id
29 + .dart_tool/
30 + .flutter-plugins-dependencies
31 + .pub-cache/
32 + .pub/
33 + /build/
34 + /coverage/
35 +
36 + # Symbolication related
```

Screenshots of the application:





Patients

Priority Alerts

 Search patient by name or ID 

Priority Alerts

 Refresh

Malaysia

3h ago

High Glucose Reading

Post-meal glucose reading of 250 mg/dL

 View Details

Sydney

Nov 5, 2:59 PM

High Blood Pressure

Blood pressure reading of 160/95 mmHg

 View Details

Malaysia

Nov 6, 2:59 PM

Low Physical Activity

No recorded physical activity for 5 consecutive days

 View Details

Germany

8h ago

High HbA1c Reading

Latest HbA1c reading of 9.2%

 View Details



Clinician Dashboard



Patients

Priority Alerts

Search patient by name or ID

Patients (5)

Malaysia ID: P001 Last Update 2h ago >
54 years, Male Type 2 Diabetes

Notifications

Malaysia 3h ago
High Glucose Reading
Post-meal glucose reading of 250 mg/dL
[View Details](#)

Sydney Nov 5, 2:59 PM
High Blood Pressure
Blood pressure reading of 160/95 mmHg
[View Details](#)

Malaysia Nov 6, 2:59 PM
Low Physical Activity
No recorded physical activity for 5 consecutive days
[View Details](#)



Clinician Profile



Dr. Example Clinician

Endocrinologist

Personal Information

Name



Dr. Example Clinician

Gender



Male



Country



Malaysia



Mobile Phone Number



+601234567890

Professional Information

Type of Profession



Endocrinologist



Account Information

Login ID



clinician01

Password



••••••••

Malaysia

Overview

Health Data

AI Chatbot

M

Malaysia

54 years, Male

Type 2 Diabetes

ID: P001

Last Update: 2h ago

+601234567890

Message

AI-Generated Health Summary

Patient shows consistent post-meal glucose spikes, particularly after lunch. Physical activity is below recommended levels. Recent HbA1c shows poor control at 8.2%. Recommend lifestyle modification and potential medication adjustment.

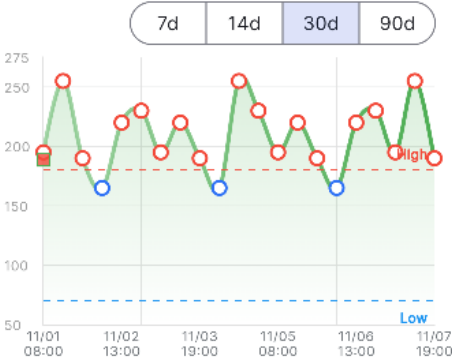
Detected Patterns

- Post-meal glucose spikes, particularly after carb-heavy lunches
- Consistently low physical activity on weekdays
- Poor medication adherence in the evening doses

AI Recommendations

- Reduce carbohydrate intake at lunch
- Incorporate 15-minute walks after meals

Glucose & HbA1c Trends



mg/dL mmol/L

Avg Glucose High Events Low Events Latest HbA1c

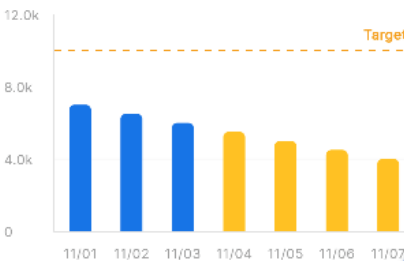
208 mg/dL

18

0

8.2%

Physical Activity



← Malaysia

Overview

Health Data

AI Chatbot

 Florence Bot

Hi, I am your AI assistant. How can I help with Malaysia's care today?

 Type your question about Malaysi...



2.4. Basil (Lead AI Engineer and Data Architect)

My work this sprint was centred on the core AI functionalities of the platform. I successfully integrated the DeepSeek-V3.1 AI model to create a functional chatbot interface. I also developed the initial logic for the personalized recommendation engine and the anomaly detection system that is also powered by DeepSeek-V3.1. A significant part of my effort was spearheading the development of algorithms for real-time pattern detection, explainable AI recommendations, and emergency detection, which form the brain of our proactive health monitoring system.

Number of commits made by Basil:

14 commits for sprint 1(main branch, darklorddad/Swinburne-COS40005-CTPA-FLORENCE) + 9 commits in sprint2(In another repository created by Alif, Harriseu/FLORENCE_Digital_Health_Platform) + 5 commits for Sprint 2 (The FYP repo)
= **total of 28 small but meaningful and complete implementation commits** to GitHub.



Figure Commits in the FYP repository(darklorddad/Swinburne-COS40005-CTPA-FLORENCE):

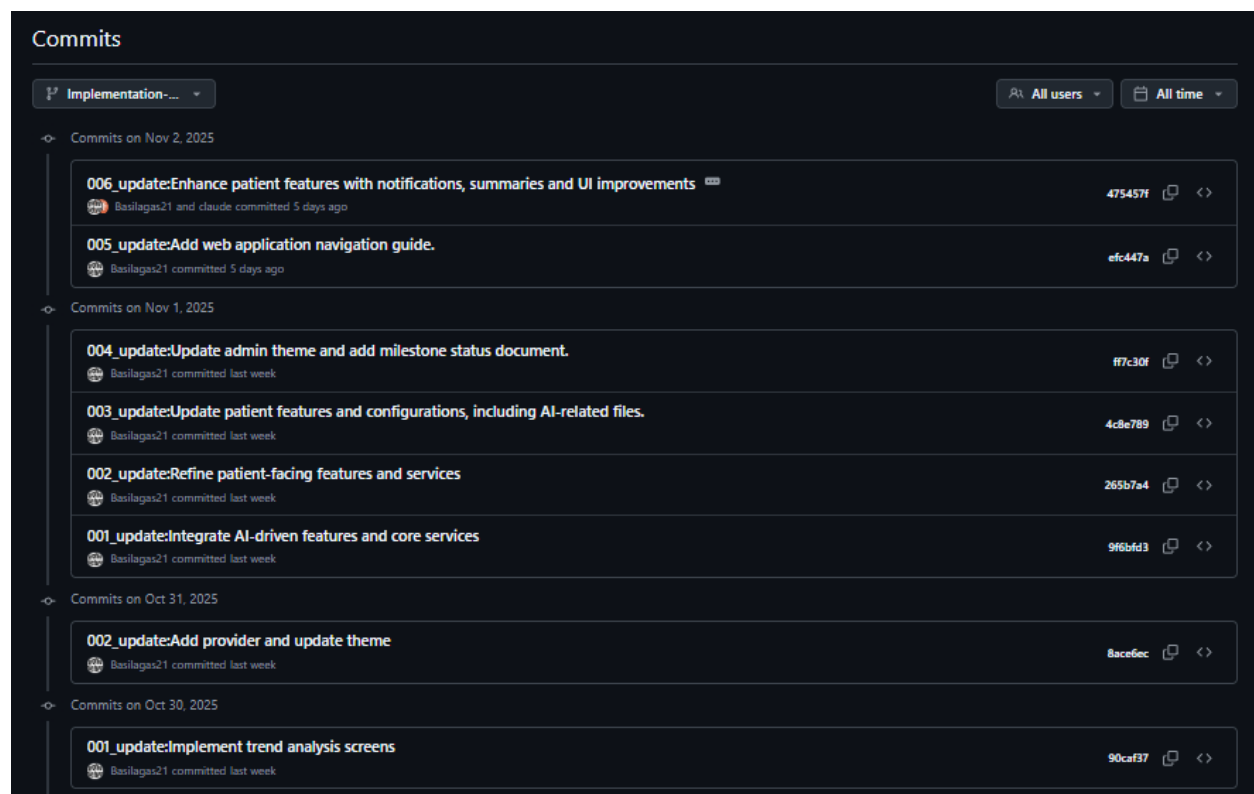


Figure Commits for sprint 2 (in Alif's repository for development and testing purposes):

Reason of separation workflow of different repository:

Progress on the main repository appeared to have slowed, and updates were not reflected until I discussed the patient-side UI design and web application implementation with Alif. It was then clarified that Alif had created a separate repository for the patient-side development and had been working there. I subsequently requested and received access to that repository to implement my AI integrations.

After completing the patient-side web application with integrated AI functionalities, both Alif and I merged our work back into the main repository managed by Daniel, which serves as the central project repository for all team members.

Following this integration, Edison and Daniel will handle most of the database connection tasks in Sprint 3. Since Alif, Ashley, and I developed our components in separate branches and workflows, we will align our implementations to ensure full

integration of the web application with the backend, following the established database schema. The connection setup is currently about 50% complete and remains in progress.

Evidence of Work

1. AI Chat Bot

For the user to interact with the AI health bot in the chat, they have to type anything they want in the text box that says “Ask me anything”. Once the user is ready to send the message, user press the blue button with the arrow to send the message.

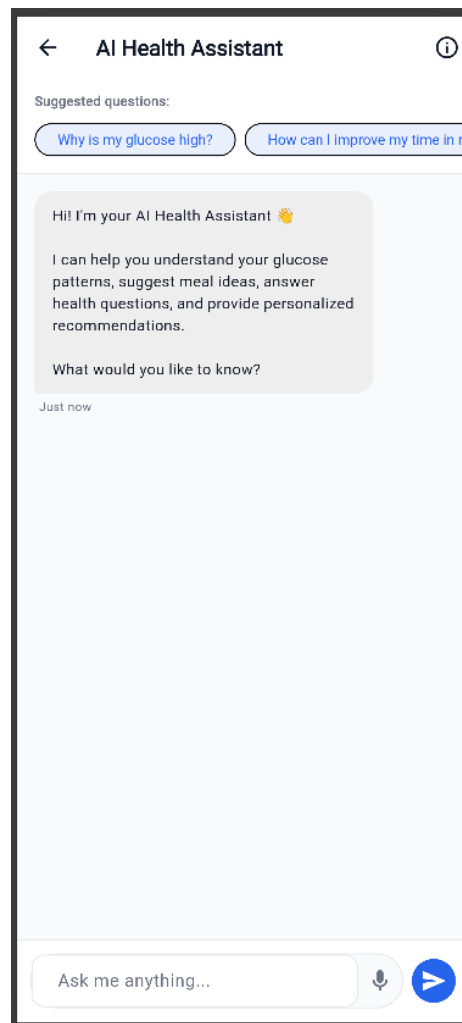


Figure Default state of AI Chatbot

Since the Ai model is integrated with DeepSeek-V3.1, it can also automatically detect the language that is sent by the user.

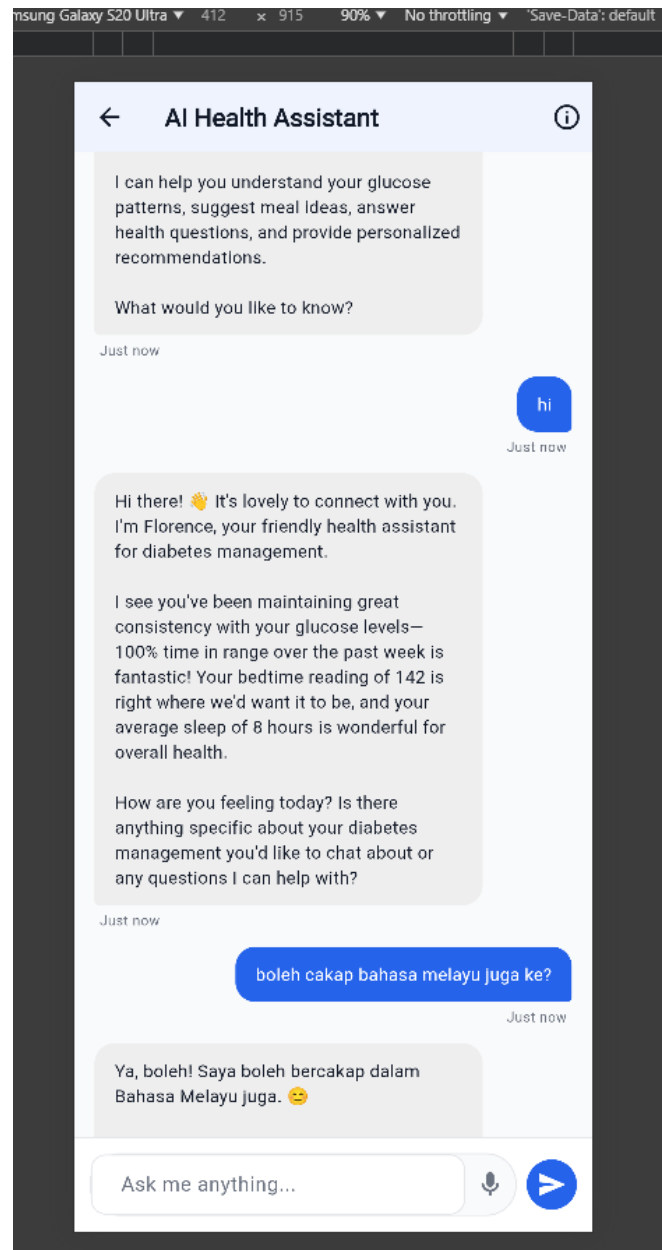


Figure AI chatbot with multi-lingual capabilities

The screenshot below show the questions that are asked by the users about their health related issue. Then, the AI health assistant will attempt to answer the users

message and reply based on the objective of the Biotective company, which is to ensure the health of their patients.

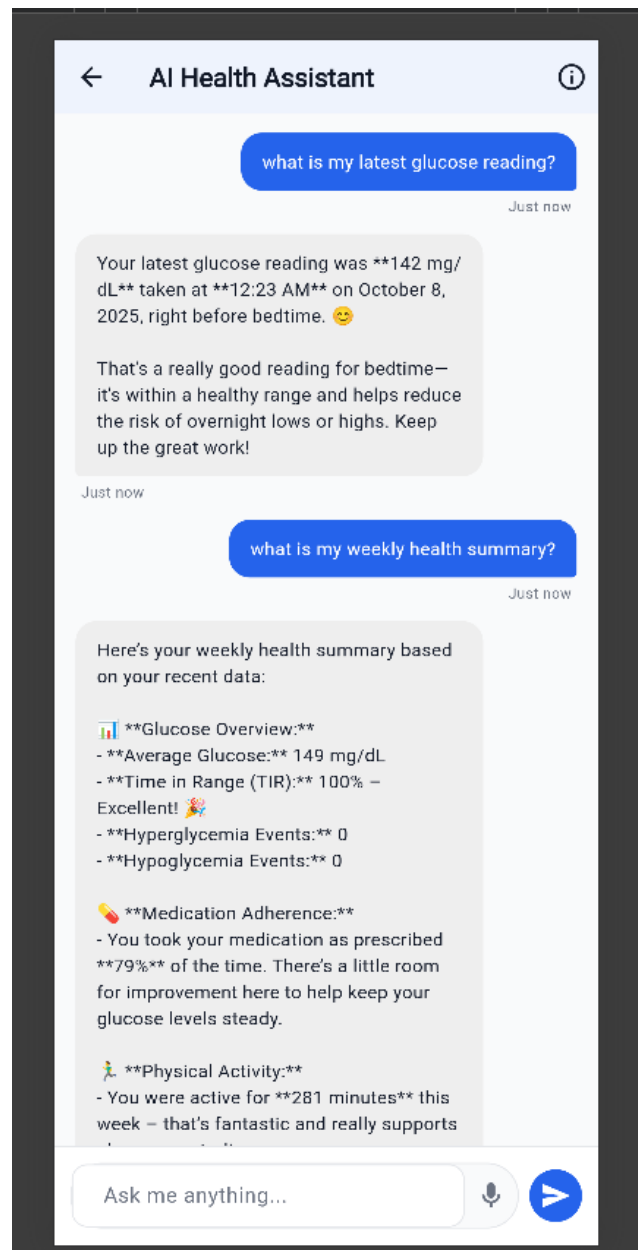


Figure AI Chatbot answering user's messages

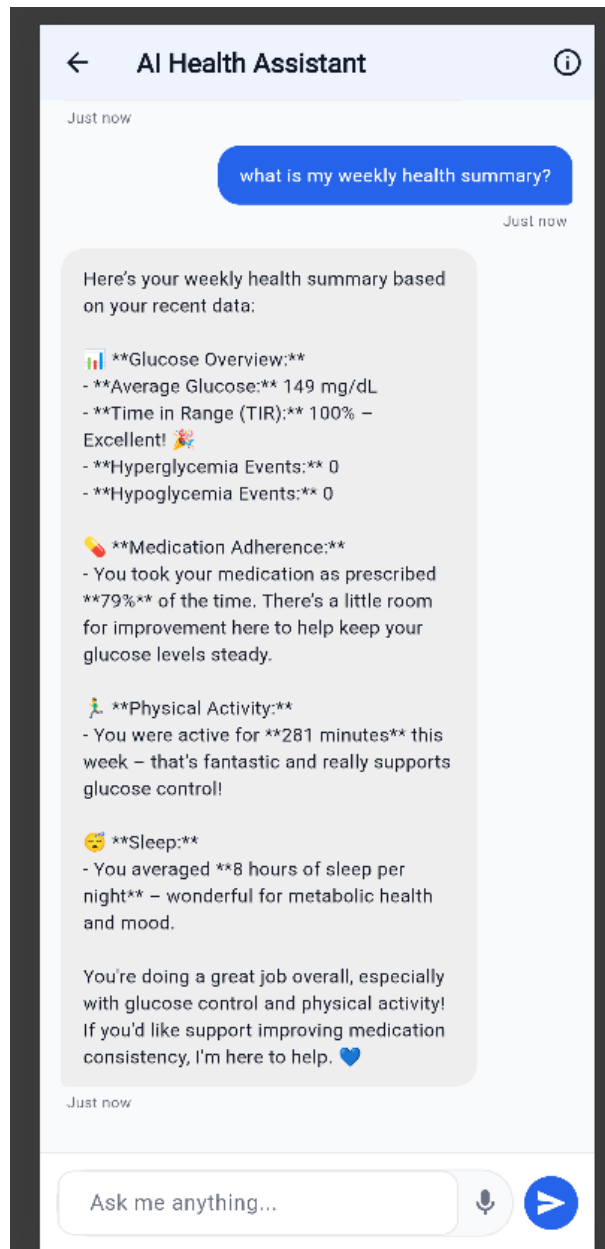


Figure AI Chatbot fetching the patients data to evaluate and respond

The screenshot below demonstrates that the DeepSeek-V3.1 AI model has been fine-tuned to align with Biotective's objectives of guiding patients in addressing their health concerns. The model is designed to assess the relevance and intent of each query to ensure it contributes effectively to assisting and supporting patients.

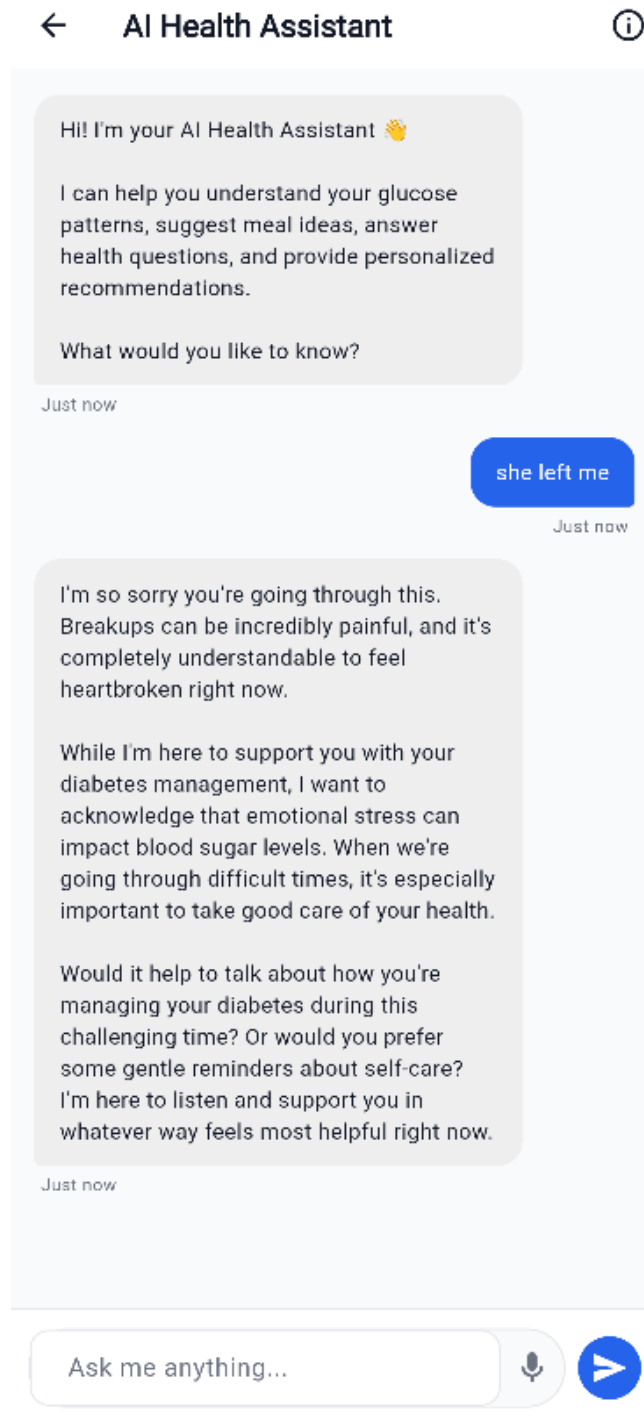


Figure Testing the Fine-tuned AI model to fulfill Biotective's objectives

001_update: Integrate AI-driven features and core services

25 changed files

- lib\core\providers\app_providers.dart
- lib\core\data_anonymization_service.dart
- lib\core\security\encryption_service.dart
- lib\core\services\deepseek_service.dart
- lib\core\pattern_detection_service.dart
- lib\core\service\notification_models.dart
- lib\core\service\notification_service.dart
- lib\core\service\schema_definitions.dart
- lib\core\supabase_health_service.dart
- lib\features\ad\risk_scoring_service.dart
- lib\features\patient\chat\services\chatbot_service.dart
- lib\features\patient\health_data_models.dart
- lib\features\patient\health_data_provider.dart
- lib\features\data_ingestion_service.dart
- lib\features\recommendation_models.dart
- lib\features\recommendation_engine.dart
- lib\features\health_summary_service.dart
- patients.csv
- test\unit\services\deepseek_service_test.dart
- test\features\recommendation_engine_test.dart
- test\patients.csv

```
45 +   });
46 + }
47 + }
48 +
49 + /// Service for AI-powered health chatbot
50 + class ChatbotService {
51 +   final DeepSeekService _deepseek = DeepSeekService();
52 +   final DataIngestionService _dataService = DataIngestionService();
53 +
54 +   // Singleton pattern
55 +   static final ChatbotService _instance = ChatbotService._internal();
56 +   factory ChatbotService() => _instance;
57 +   ChatbotService._internal();
58 +
59 +   // Conversation history
60 +   final List<ChatMessageModel> _conversationHistory = [];
61 +
62 +   List<ChatMessageModel> get conversationHistory =>
63 +     List.unmodifiable(_conversationHistory);
64 +
65 +   /// Send a message to the chatbot
66 +   Future<ChatMessageModel> sendMessage(String userMessage) async {
67 +     // Add user message to history
68 +     final userChatMessage = ChatMessageModel(
69 +       id: 'msg_${DateTime.now().millisecondsSinceEpoch}_user',
70 +       role: 'user',
71 +       content: userMessage,
72 +       timestamp: DateTime.now(),
73 +     );
```

Figure Chatbot services

```
001_update: Integrate AI-driven features and core services
Basilagas21 9f6bfd3 +7986 -0

25 changed files
lib\core\services\ai\deepseek_service.dart
lib\core\config\environment.dart
lib\core\providers\app_providers.dart
lib\core\sec...\data_anonymization_service.dart
lib\core\security\encryption_service.dart
lib\core\services\ai\deepseek_service.dart
lib\core\service...\pattern_detection_service.dart
lib\core\services\not...\notification_models.dart
lib\core\services\noti...\notification_service.dart
lib\core\services\sup...\schema_definitions.dart
lib\core\services...supabase_health_service.dart
lib\features\admin\p...\risk_scoring_service.dart
lib\features\patient\chat...\chatbot_service.dart
lib\features\patient...\health_data_models.dart
lib\features\patient...\health_data_provider.dart
lib\features\patie...\data_ingestion_service.dart
lib\features\pa...\recommendation_models.dart
lib\features\pat...\recommendation_engine.dart
lib\features\pati...\health_summary_service.dart

249 + }
250 +
251 + /// Chatbot conversation
252 + Future<String> chatbot({
253 +   required String userMessage,
254 +   required Map<String, dynamic> healthContext,
255 +   List<ChatMessage>? conversationHistory,
256 + }) async {
257 +   final systemPrompt = '''
258 +   You are FLORENCE, a friendly AI health assistant for diabetes management.
259 +
260 +   Patient's recent health context:
261 +   ${_formatHealthContext(healthContext)}
262 +
263 +   Your role:
264 +   - Answer questions about their health data
265 +   - Provide guidance and support
266 +   - Explain diabetes management concepts
267 +   - Offer personalized tips based on their data
268 +   - Be warm, encouraging, and non-judgmental
269 +
270 +   Important:
271 +   - Never diagnose or provide medical advice
272 +   - Encourage them to consult healthcare providers for concerns
273 +   - Reference their actual data when relevant
274 +   - Keep responses concise and clear
275 +   ''';
276 +
277 +   final messages = <ChatMessage>[
```

Figure Using DeepSeek's services to power the AI chatbot

2. AI powered insights

For the AI-insights, the information displayed are automatically generated when the patient's data are updated in the database. The implementation of the AI-insights exists in some of the pages of the webapp, which are the main dashboard, trends visualisation, Glucose trend details, pattern detection (in each of the pattern cards when generated) and in the weekly summary pages(for each week).

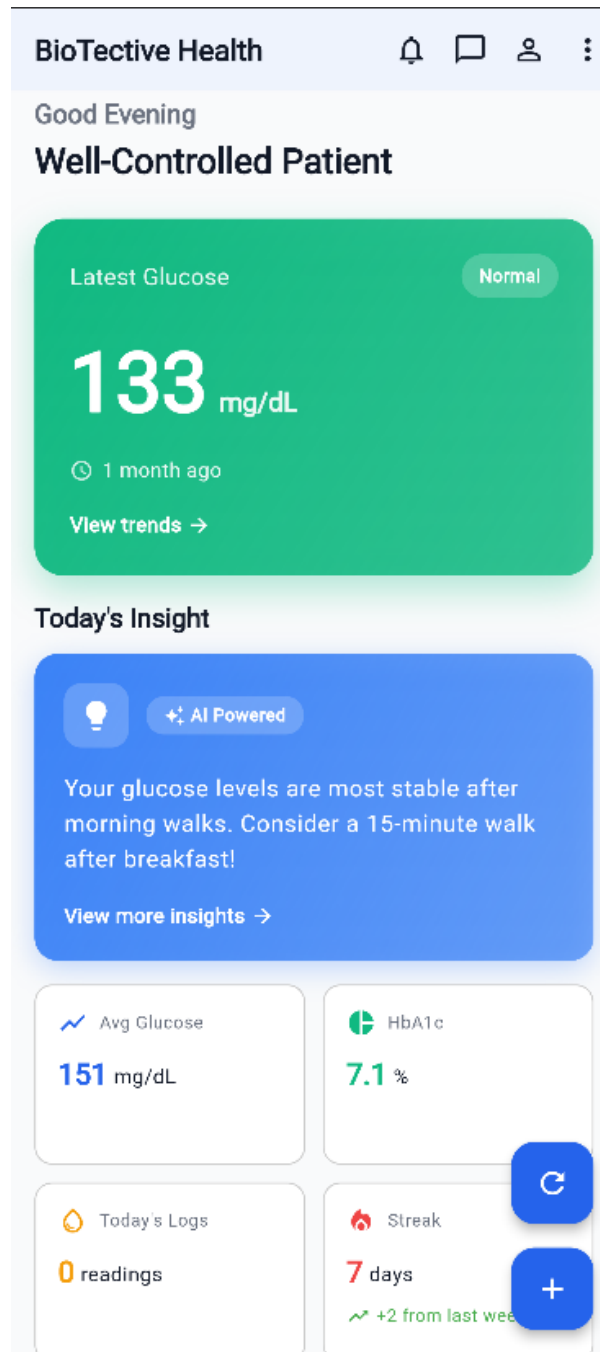


Figure AI insight in the patient's main dashboard

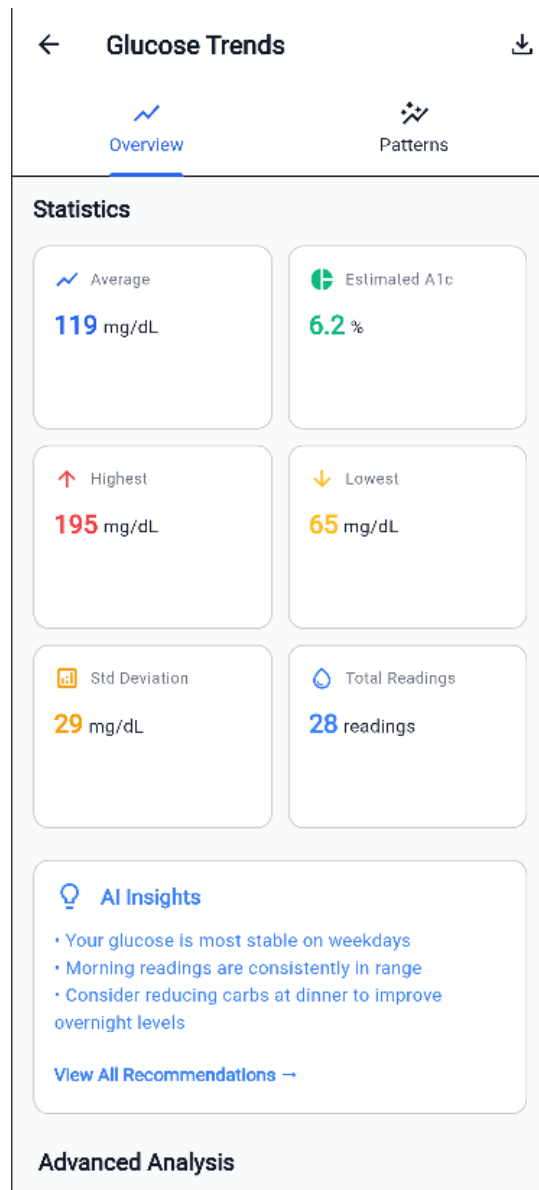


Figure AI insight in the trend visualisation

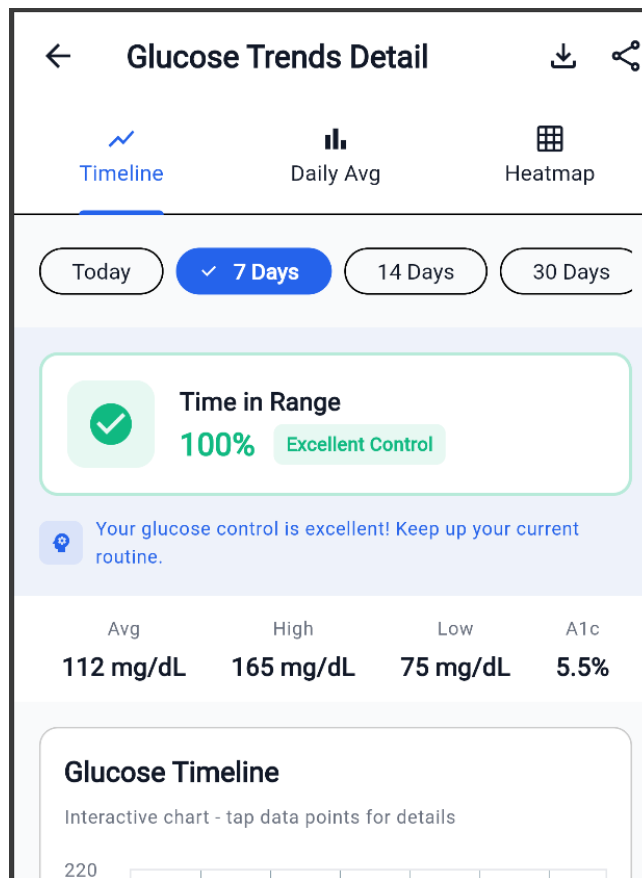


Figure AI insight in the Glucoes trend details

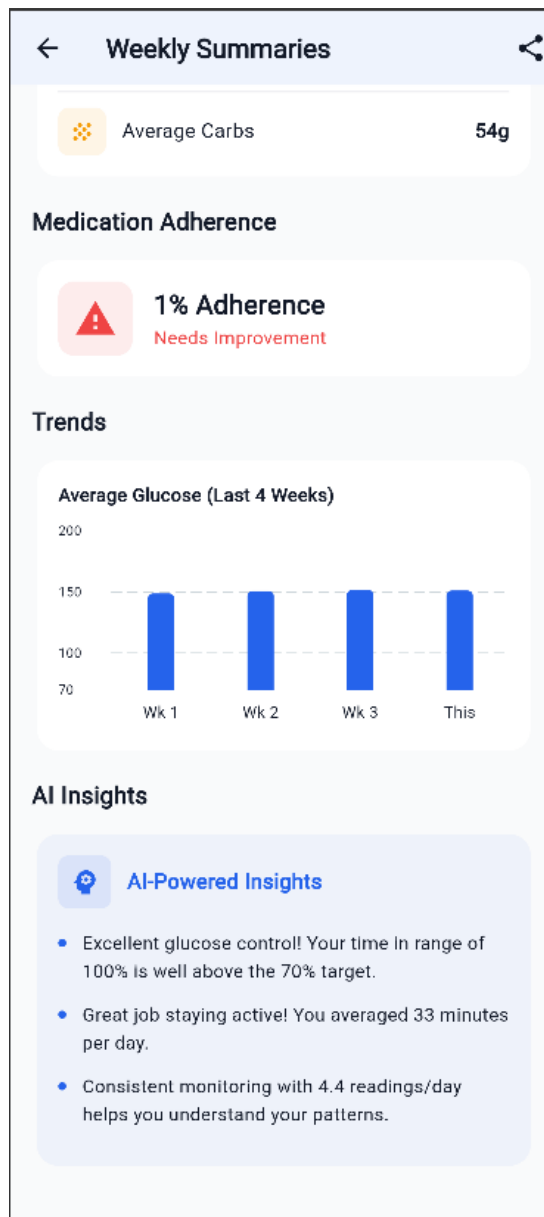


Figure AI insight in the Weekly Summaries

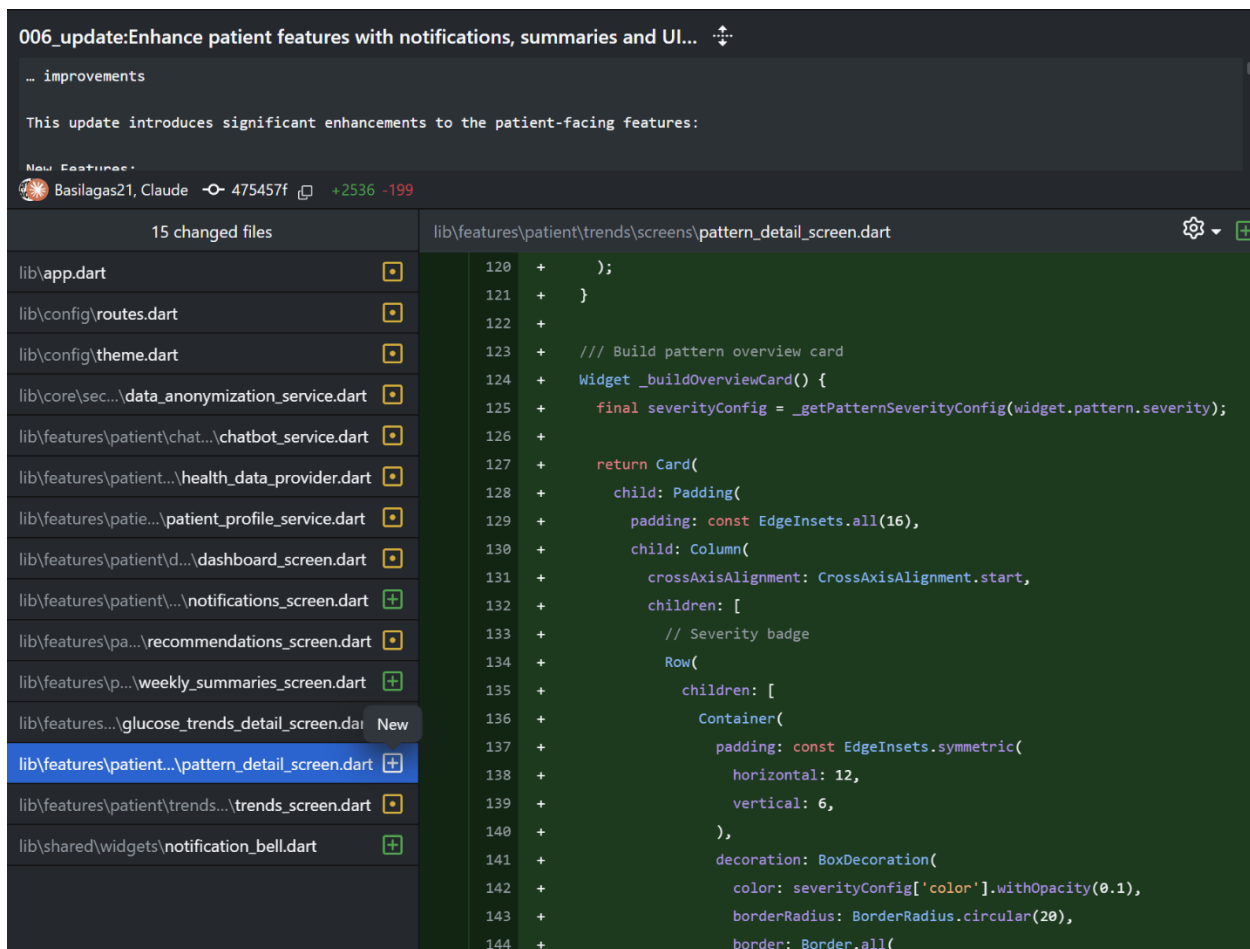


Figure The implementation of AI insights in the Patient's detail screen

006_update:Enhance patient features with notifications, summaries and UI...
... improvements

This update introduces significant enhancements to the patient-facing features:

New Estimates:
Basilagas21, Claude 475457f +2536 -199

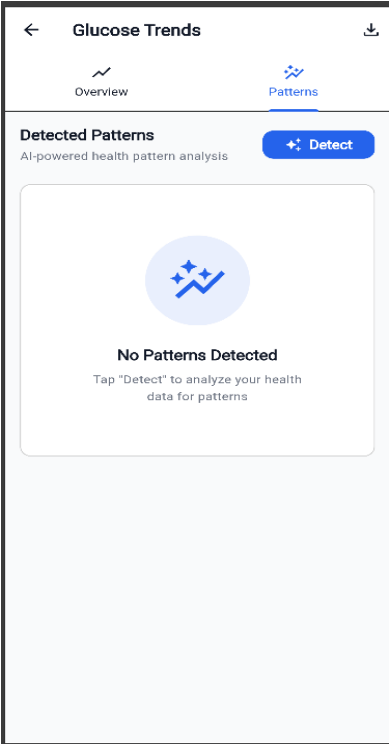
15 changed files

lib\app.dart
lib\config\routes.dart
lib\config\theme.dart
lib\core\sec...data_anonymization_service.dart
lib\features\patient\chat...chatbot_service.dart
lib\features\patient...health_data_provider.dart
lib\features\patie...patient_profile_service.dart
lib\features\patient\d...dashboard_screen.dart
lib\features\patient\...notifications_screen.dart
lib\features\pa...recommendations_screen.dart
lib\features\p...weekly_summaries_screen.dart
lib\features...glucose_trends_detail_screen.dart
lib\features\patient...pattern_detail_screen.dart
lib\features\patient\trends...trends_screen.dart
lib\shared\widgets\notification_bell.dart

lib\features\pat...weekly_summaries_screen.dart
539 +
540 + const SizedBox(width: 12),
541 +
542 + const Text(
543 + 'AI-Powered Insights',
544 + style: TextStyle(
545 + fontSize: 16,
546 + fontWeight: FontWeight.bold,
547 + color: AppTheme.primaryBlue,
548 +),
549 +),
550 +
551 + const SizedBox(height: 16),
552 + ...insights.map((insight) => Padding(
553 + padding: const EdgeInsets.only(bottom: 12),
554 + child: Row(
555 + crossAxisAlignment: CrossAxisAlignment.start,
556 + children: [
557 + Container(
558 + width: 6,
559 + height: 6,
560 + margin: const EdgeInsets.only(top: 7, right: 12),
561 + decoration: const BoxDecoration(
562 + color: AppTheme.primaryBlue,
563 + shape: BoxShape.circle

Figure The implementation of AI insights in the weekly summaries

3. AI-powered Patient's Data Pattern Recognition



The user can click to the blue button that says “☀ Detect” to have the AI model to analyze the patterns based on the patients data which it will refer to the database. Then, the AI model will evaluate based on the data points that it selected. After the detected patterns are finished, one or more cards will be generate based on the priority of the recommendation for the user is as shown in the screenshot below:

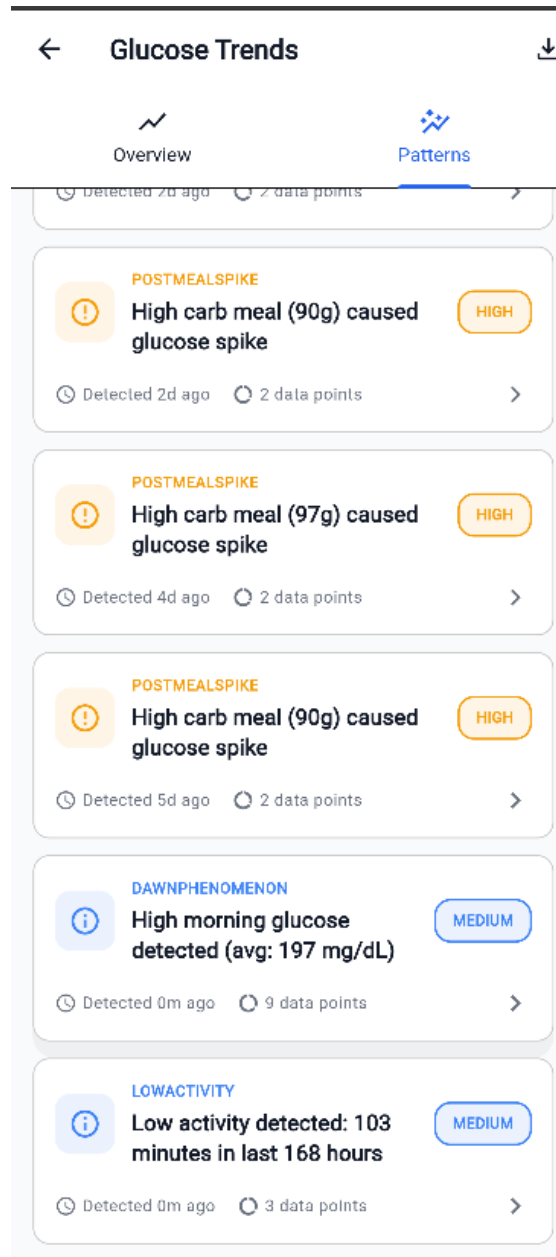


Figure AI-powered Pattern Recognition and AI-generated pattern cards

Once the pattern cards are generated, the user can also choose to click on one of the cards and view the details of the AI-generated pattern that are evaluated based on the patient's data points. Also, inside each of the AI-generated pattern card, there is also AI-powered insights. The screenshot below shows the page when the user clicks one of the pattern cards:

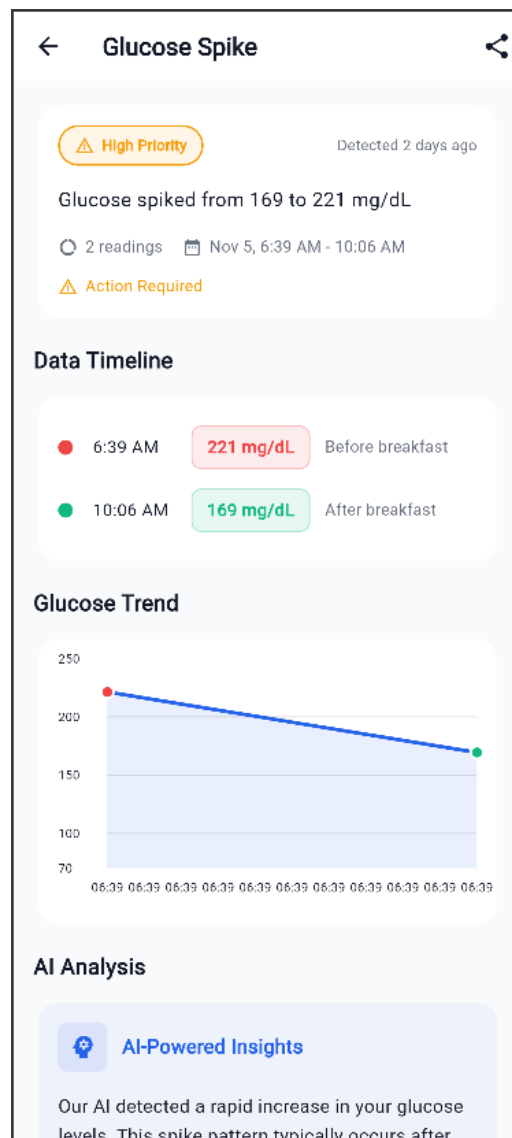


Figure The details of one of the pattern card generate top page

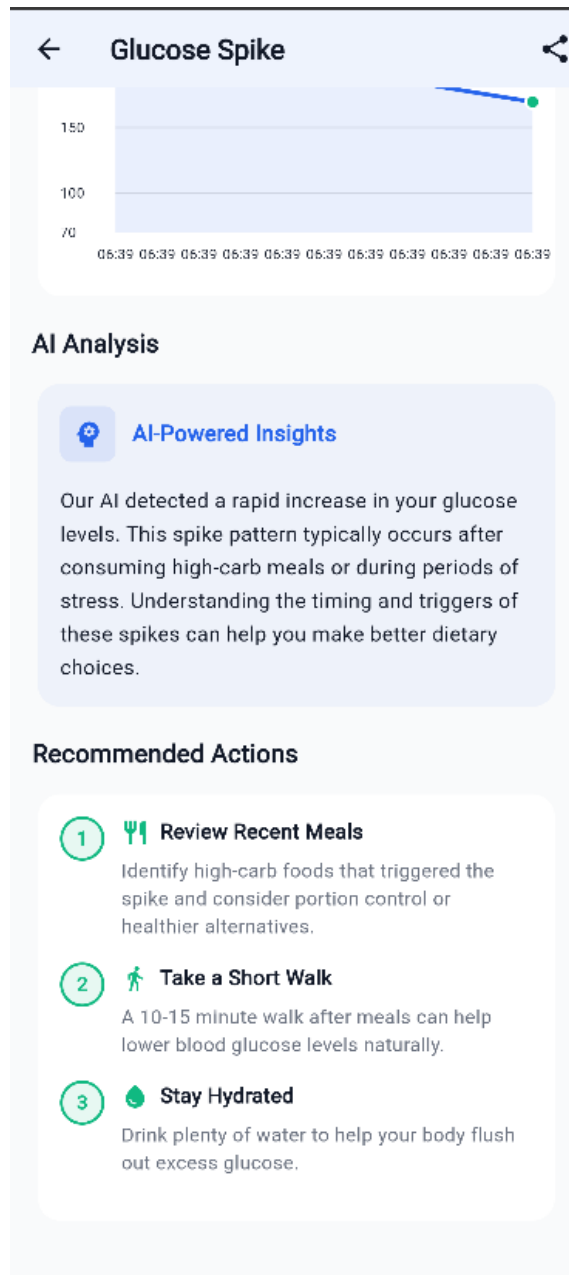


Figure The details of one of the pattern card generate bottom page

Normal pattern recognition:

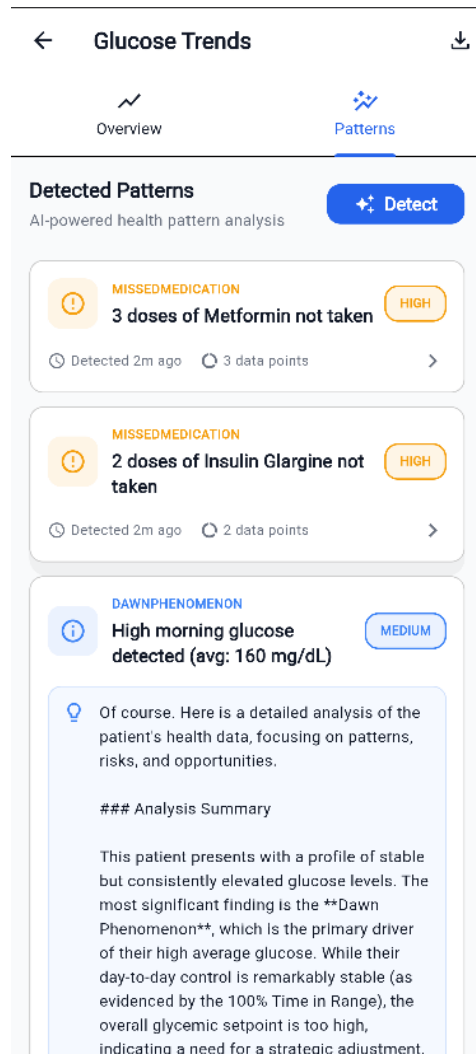


Figure Pattern detection for normal patients

Patient-at-risk Detection:

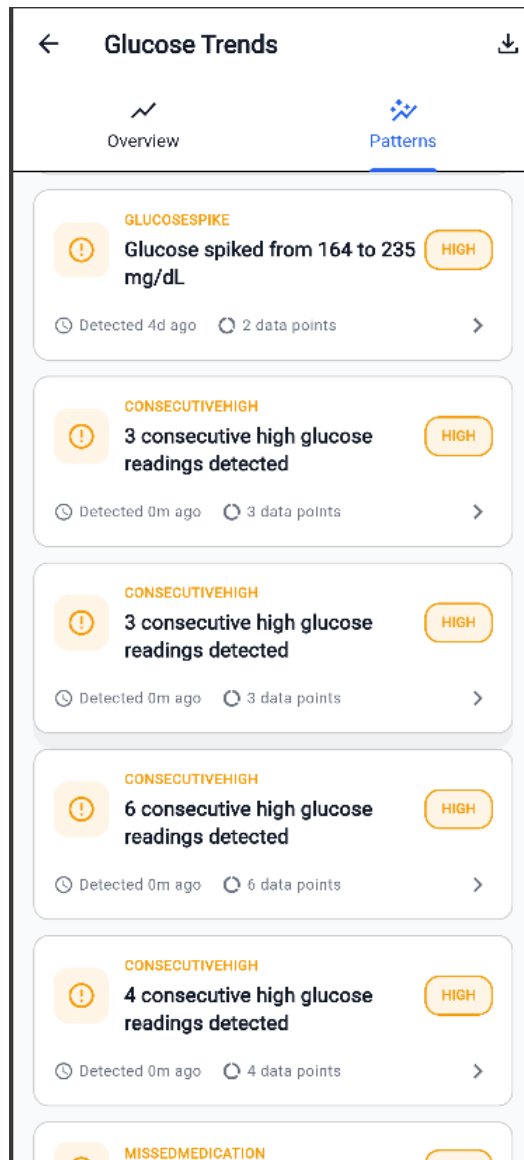


Figure Pattern detection of high-risk patient

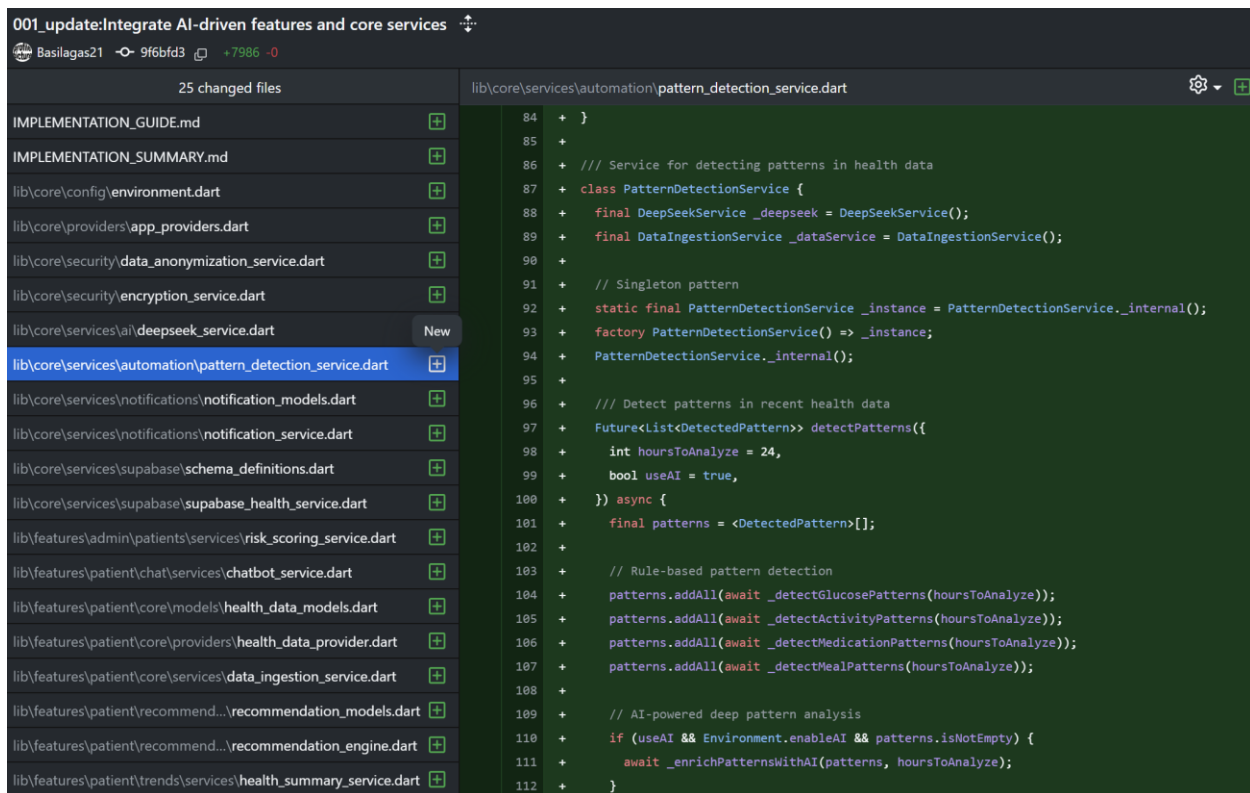


Figure Implementation of the AI-powered pattern recognition

4. AI-Recommendation

As for the AI-recommendation page, there are three tabs. Active, Completed, and Dismissed. In the Active tab, recommendations are AI-generated based on what the patient should focus on. The patient can also generate the new recommendations. As of now the progress done in Sprint 2, the user can keep generating five new recommendations every time the user press the blue button, “Generate New Recommendations”. This occurs because the database connection is still being established and a work in progress. When the user is done by fulfilling the recommendation, the can press the green outlined button that says “Mark Done” and they choose to ignore the recommendation, they can press the “Dismiss”. Below is a screenshot of the Ai-generated recommendations.

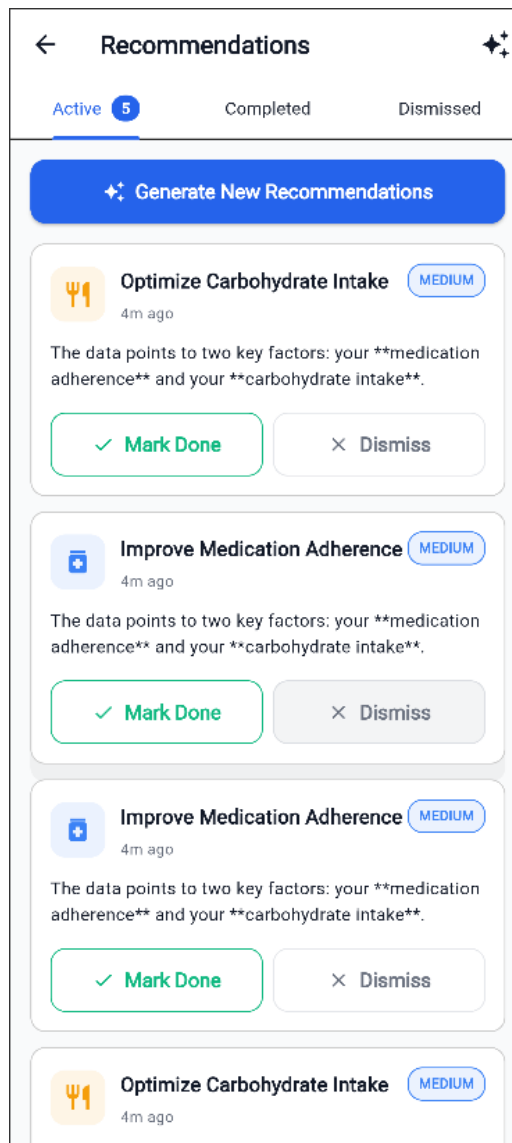


Figure The AI-powered recommendation in the Active tab

The user can also click into one of the recommendation card to view a more in-depth detailed about the generated recommendation. The details about the recommendation card are also AI generated. For example, as shown in one of the recommendation card which is the "Improve Medication Adherence", the "Why This Matters" section as show in the screenshot below is generated by AI that is evaluated based on the chosen data points. In this case, the chosen data points should related to glucose to improve medication adherence. As the user scroll down more, there are Action

Steps section for the AI to suggest the user on what kind of actions to take based on the recommendation generated. The steps for the action to take are also AI generated.

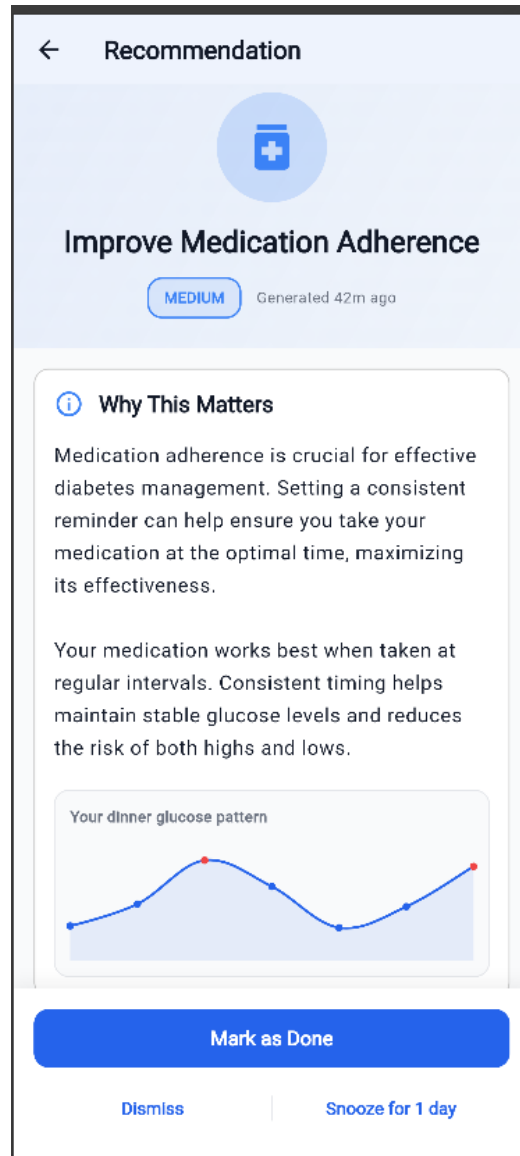


Figure Details of one of the recommendation top page

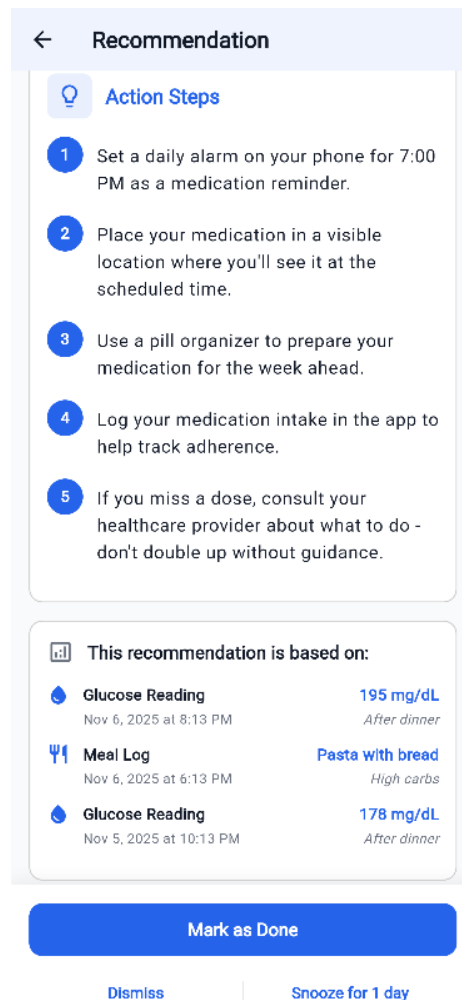


Figure Details of one of the recommendation bottom page

001_update: Integrate AI-driven features and core services

Basilagas21 9f6bfd3 +7986 -0

25 changed files

lib/features/patient/recommendations/models/recommendation_models.dart

lib\core\security\encryption_service.dart

lib\core\services\ai\deepseek_service.dart

lib\core\services\automation\pattern_detection_service.dart

lib\core\services\notifications\notification_models.dart

lib\core\services\notifications\notification_service.dart

lib\core\services\supabase\schema_definitions.dart

lib\core\services\supabase\supabase_health_service.dart

lib\features\admin\patients\services\risk_scoring_service.dart

lib\features\patient\chat\services\chatbot_service.dart

lib\features\patient\core\models\health_data_models.dart

lib\features\patient\core\providers\health_data_provider.dart

lib\features\patient\core\services\data_ingestion_service.dart

lib\features\patient\recommend...\recommendation_models.dart

lib\features\patient\recommend...\recommendation_engine.dart

lib\features\patient\trends\services\health_summary_service.dart

patients.csv

test\unit\services\deepseek_service_test.dart

test\unit\services\recommendation_engine_test.dart

test_patients.csv

tools\patient_generator.dart

220 + /// Data point that triggered a recommendation

221 + @immutable

222 + class DataPoint {

223 + final String type; // e.g., "glucose_reading", "meal", "activity"

224 + final String description; // Human-readable description

225 + final String value; // The actual value

226 + final DateTime timestamp;

227 +

228 + const DataPoint({

229 + required this.type,

230 + required this.description,

231 + required this.value,

232 + required this.timestamp,

233 + });

234 +

235 + Map<String, dynamic> toJson() {

236 + return {

237 + 'type': type,

238 + 'description': description,

239 + 'value': value,

240 + 'timestamp': timestamp.toIso8601String(),

241 + };

242 + }

243 +

244 + factory DataPoint.fromJson(Map<String, dynamic> json) {

245 + return DataPoint(

246 + type: json['type'] as String,

247 + description: json['description'] as String,

248 + value: json['value'] as String,

Figure implementation of the Recommendation for Data points fetching

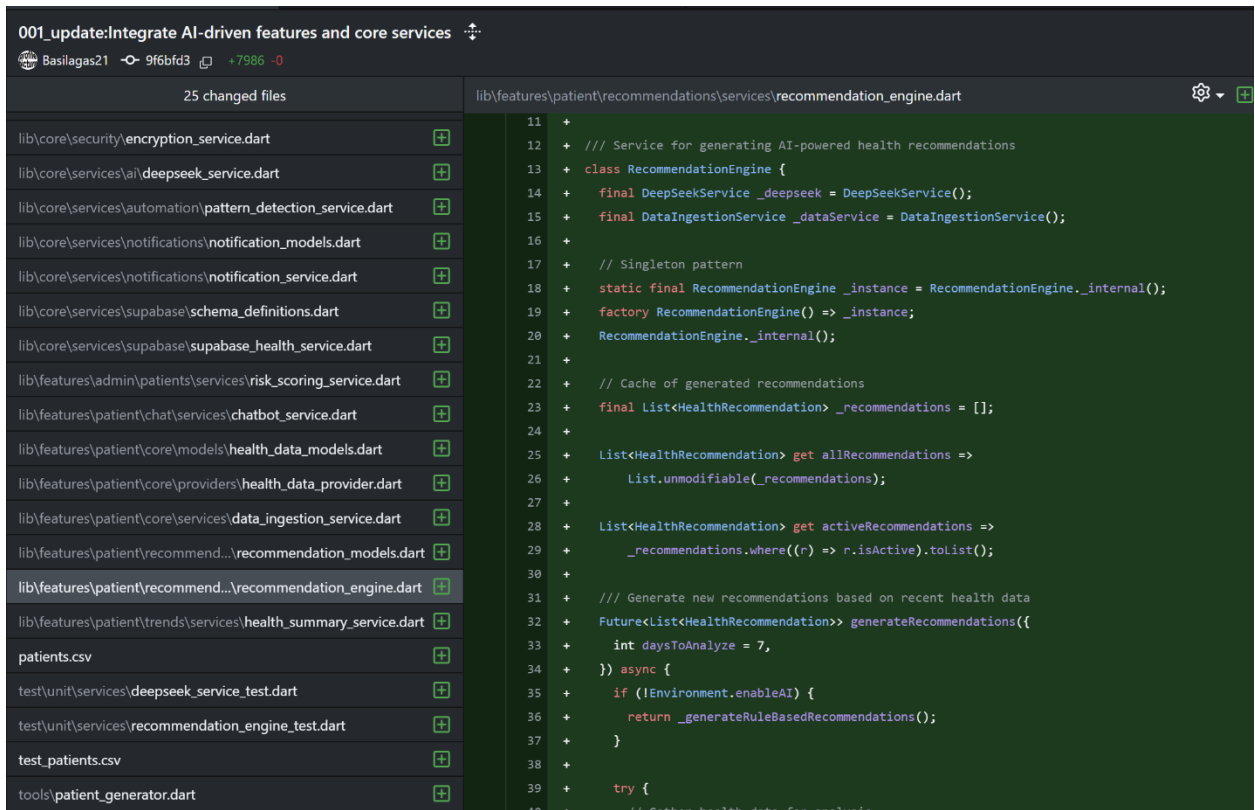


Figure Implementation of the Recommendation with AI generation

001_update: Integrate AI-driven features and core services

Basilagas21 9f6bfd3 +7986 -0

25 changed files

lib/features/patient/recommendations/models/recommendation_models.dart

lib\core\security\encryption_service.dart

lib\core\services\ai\deepseek_service.dart

lib\core\services\automation\pattern_detection_service.dart

lib\core\services\notifications\notification_models.dart

lib\core\services\notifications\notification_service.dart

lib\core\services\supabase\schema_definitions.dart

lib\core\services\supabase\supabase_health_service.dart

lib\features\admin\patients\services\risk_scoring_service.dart

lib\features\patient\chat\services\chatbot_service.dart

lib\features\patient\core\models\health_data_models.dart

lib\features\patient\core\providers\health_data_provider.dart

lib\features\patient\core\services\data_ingestion_service.dart

lib\features\patient\recommen... recommendation_models.dart

lib\features\patient\recommen... recommendation_engine.dart

lib\features\patient\trends\services\health_summary_service.dart

patients.csv

test\unit\services\deepseek_service_test.dart

test\unit\services\recommendation_engine_test.dart

test_patients.csv

tools\patient_generator.dart

254 + /// Recommendation generation context

255 + class RecommendationContext {

256 + final List<Map<String, dynamic>> glucoseReadings;

257 + final List<Map<String, dynamic>> meals;

258 + final List<Map<String, dynamic>> activities;

259 + final Map<String, dynamic> summary;

260 + final DateTime analysisStart;

261 + final DateTime analysisEnd;

262 +

263 + RecommendationContext({

264 + required this.glucoseReadings,

265 + required this.meals,

266 + required this.activities,

267 + required this.summary,

268 + required this.analysisStart,

269 + required this.analysisEnd,

270 + });

271 +

272 + Map<String, dynamic> toJson() {

273 + return {

274 + 'glucoseReadings': glucoseReadings,

275 + 'meals': meals,

276 + 'activities': activities,

277 + 'summary': summary,

278 + 'analysisStart': analysisStart.toIso8601String(),

279 + 'analysisEnd': analysisEnd.toIso8601String(),

280 + };

281 + }

282 + }

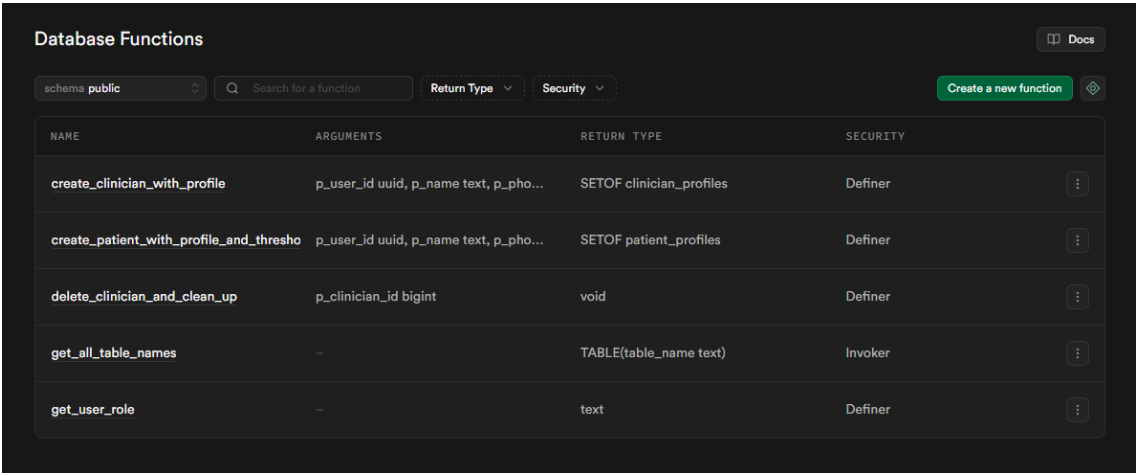
Figure Recommendation implementation with generating context

2.5. Edison (Backend Developer)

My contribution this sprint was to establish the project's robust backend infrastructure. I implemented the full user authentication system using Supabase, which handles the complete registration and login lifecycle for all user roles. I also developed a comprehensive set of RESTful API endpoints in FastAPI to manage all system entities, including patients, clinicians, and admins. To ensure the quality and stability of the backend, I set up the automated testing framework using Pytest and wrote the initial test suites for the authentication and core API endpoints.

I set up the Row-Level Security (RLS) policies in Supabase to ensure that users can only access their own data. For performance and data integrity, I implemented crucial database functions and indexes. I worked in close collaboration with the project lead, Daniel, to ensure the backend architecture aligned with the frontend and deployment requirements, providing the foundational logic upon which the API endpoints are built.

Evidence of Work



The screenshot displays the Supabase 'Database Functions' management interface. At the top, there's a header with 'Database Functions' and a 'Docs' link. Below this is a navigation bar with a schema dropdown set to 'public', a search bar, and filters for 'Return Type' and 'Security'. A 'Create a new function' button is on the right. The main area is a table listing functions with columns for Name, Arguments, Return Type, and Security. Each row has a three-dot menu on the right.

NAME	ARGUMENTS	RETURN TYPE	SECURITY
create_clinician_with_profile	p_user_id uuid, p_name text, p_phono...	SETOF clinician_profiles	Definer
create_patient_with_profile_and_thresho	p_user_id uuid, p_name text, p_phono...	SETOF patient_profiles	Definer
delete_clinician_and_clean_up	p_clinician_id bigint	void	Definer
get_all_table_names	-	TABLE(table_name text)	Invoker
get_user_role	-	text	Definer

Database Indexes

Improve query performance against your database

Docs

Index Advisor

schema public

Search for an index

Create index

SCHEMA	TABLE	NAME		
public	admin_profiles	admin_profiles_pkey	View definition	
public	admin_profiles	admin_profiles_user_id_key	View definition	
public	clinician_notes	clinician_notes_pkey	View definition	
public	clinician_profiles	clinician_profiles_pkey	View definition	
public	clinician_profiles	clinician_profiles_user_id_key	View definition	
public	daily_patient_logs	daily_patient_logs_patient_id_log_date_meal_time_key	View definition	
public	daily_patient_logs	daily_patient_logs_pkey	View definition	
public	patient_monitor_data	idx_patient_data_measured_at	View definition	
public	daily_patient_logs	idx_patient_logs_date	View definition	
public	organisations	organisations_pkey	View definition	
public	patient_monitor_data	patient_monitor_data_pkey	View definition	
public	patient_profiles	patient_profiles_pkey	View definition	

Policies

Manage Row Level Security policies for your tables

Docs

Filter tables and policies

schema public

admin_profiles

Disable RLS

Create policy

NAME	COMMAND	APPLIED TO	
Admins can manage all admin profiles	ALL	authenticated	

clinician_notes

Disable RLS

Create policy

NAME	COMMAND	APPLIED TO	
Admins can manage all clinician notes	ALL	authenticated	
Clinicians can manage notes for their patients	ALL	authenticated	

clinician_profiles

Disable RLS

Create policy

NAME	COMMAND	APPLIED TO	
Admins can manage all clinician profiles	ALL	authenticated	
Clinicians can see their own profile	SELECT	authenticated	
Clinicians can update their own profile	UPDATE	authenticated	

- ▼ data_service
 - > __pycache__
 - > .pytest_cache
 - ▼ core
 - > __pycache__
 - ✚ dependencies.py
 - ✚ utils.py
 - > data_service_devtool
 - ▼ routers
 - > __pycache__
 - ✚ admin.py
 - ✚ authentication.py
 - ✚ clinicians.py
 - ✚ patients.py
 - ▼ tests
 - > __pycache__
 - ✚ manual_admin_login_gui.py
 - ✚ test_public_auth.py
 - ✚ client.py
 - 📄 database_schema.sql
 - ✚ main.py
 - ✚ models.py
 - 📄 pytest.ini
 - 📄 requirements-dev.txt
 - 📄 requirements.txt

3. Sprint Plan

The team entered Sprint 2 with a plan to focus on **EPIC 2: AI Core and Automation**, with a backlog focused on developing the LangChain agent (T3.1), the recommendation engine (T3.2) and the automation layer (T4.1, T4.2). Our initial plan is shown below.

Story ID	User Story	Priority	Story Points
T3.1	As an AI Engineer, I want to build a LangChain agent that can reason about patient data and use tools to fetch additional information from the database.	High	21
T3.2	As a patient, I want to receive personalised recommendations (e.g., meal substitutions, activity suggestions) based on my data patterns.	High	13
T4.1	As a developer, I want to create an automation layer (LAM Triggers) that runs periodically to check for abnormal health patterns in the database.	High	8
T4.2	As a patient, I want to receive a timely reminder (e.g., push notification) to take a short walk after a high-carb meal is detected.	Medium	5

Table 2: Sprint Product Backlog

However, early in the sprint, a critical architectural impediment was identified that required an immediate pivot. The scattered nature of our multiple repositories was a direct path to realising Risk R4 - Integration Issues (Team and Project Plan, section 5) and was already hindering development. The primary goal for the sprint thus evolved to establish a complete technical architecture that was scalable and enabled parallel workflows, while still making progress on the AI backlog items. The final sprint backlog, reflecting this pivot, is shown in Figure 2.

Swinburne-COS40005-CTPA-FLORENCE

Product Backlog Task Board Project Roadmap New view

Filter by keyword or by field Discard Save

ID	Title	Status	Priority	Story points	Sprint
2	T1.1 As a developer, I want to set up the project structure with a GitHub repo and CI/CD pipeline so that we have a foundation for collaborative work. #16	Done	High		3 Sprint 1
3	T1.2 As a developer, I want to set up the Supabase project with a complete database schema so that we can store user and health data. #17	Done	High		5 Sprint 1
4	T1.3 As a patient/clinician, I want to securely sign up, log in and log out so that I can access the platform and my data is protected. #18	Done	High		8 Sprint 1
5	T1.4 As a developer, I need a Python script to simulate and populate the database with realistic patient data so that we can develop and test features. #19	Done	High		8 Sprint 1
6	T2.1 As a patient, I want to see a dashboard with visualisations of my key health metrics (glucose, HbA1c, diet, activity) so that I can understand my trends. #20	Done	High		13 Sprint 1
7	EPIC 2: AI Core & Automation (M3 & M4) #30	Epic			
8	T3.1 As an AI Engineer, I want to build a LangChain agent that can reason about patient data and use tools to fetch additional information from the database. #21	In Progress	High		21 Sprint 2
9	T3.2 As a patient, I want to receive personalised recommendations (e.g., meal substitutions, activity suggestions) based on my data patterns. #22	In Progress	High		13 Sprint 2
10	T4.1 As a developer, I want to create an automation layer (IAM Triggers) that runs periodically to check for abnormal health patterns in the database. #23	Backlog	High		8 Sprint 2
11	T4.2 As a patient, I want to receive a timely reminder (e.g., push notification) to take a short walk after a high-carb meal is detected. #24	Backlog	Medium		5 Sprint 2
12	EPIC 3: Clinician View & Finalisation (M5 & M6) #31	Epic			
13	T5.1 As a clinician, I want to see a dashboard with a list of my assigned patients, highlighting those who need urgent attention. #25	Backlog	High		13 Sprint 3
14	T5.2 As a patient, I want to use a chatbot to ask simple questions about my health data (e.g., "What was my average glucose this week?"). #26	Backlog	Medium		13 Sprint 3
15	T6.1 As a developer, I want to conduct end-to-end system testing using a multi-patient dataset to ensure all components function correctly together. #27	Backlog	High		8 Sprint 3
16	T6.2 As a team, we want to finalise all project documentation, including a comprehensive user guide and technical documentation, for final submission. #28	Backlog	High		13 Sprint 3
17	Miscellaneous #34	Miscellaneous			
18	Architectural Refactor to Monorepo #33	Done			

Figure 2: Complete Product Backlog (GitHub table view)

4. Sprint Progress

The team's progress against the overall project roadmap was tracked using GitHub's Roadmap and Board views. The most significant action this sprint was the strategic pivot to perform a complete architectural refactoring. This was identified as a critical, high-priority task that superseded some of the planned feature work. The project was successfully restructured from multiple disparate repositories into a unified monorepo with a microservices-style architecture. This structure now allows for fully independent, parallel development. As a result, tasks T4.1 and T4.2 were formally moved to the next sprint.

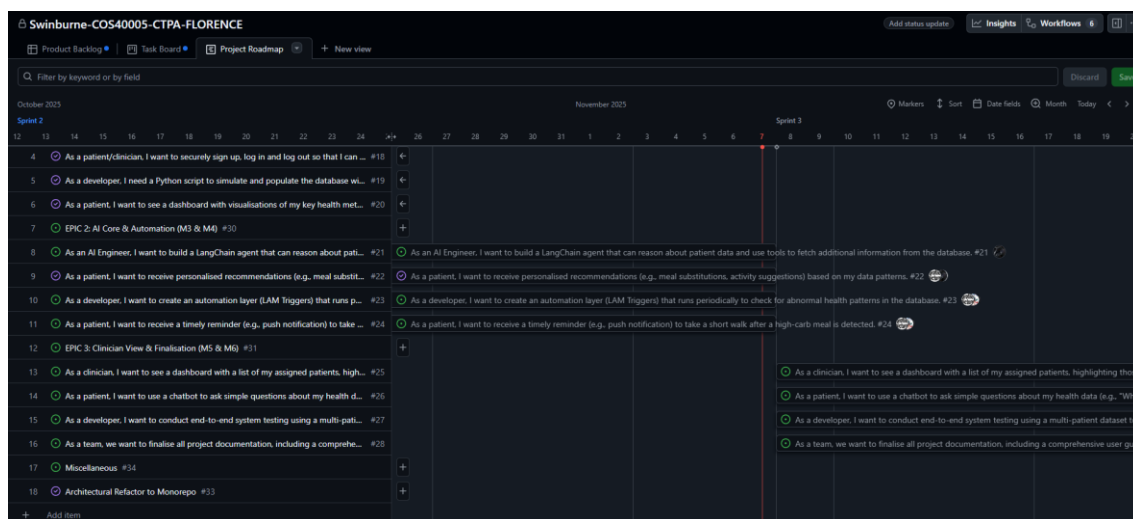


Figure 3: Project Roadmap

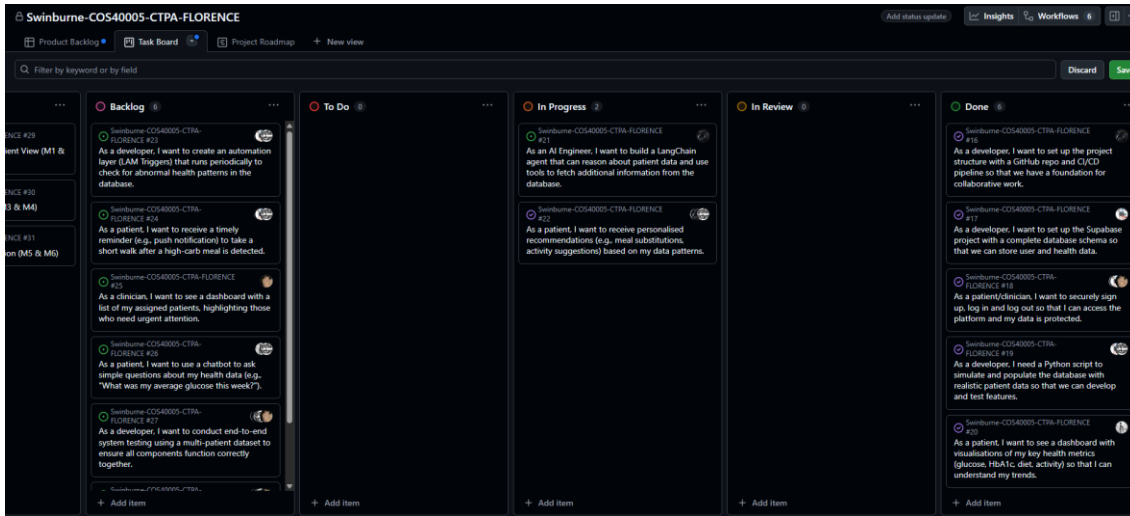


Figure 4: Sprint Task Board



Figure 5: The number of commits in the repository

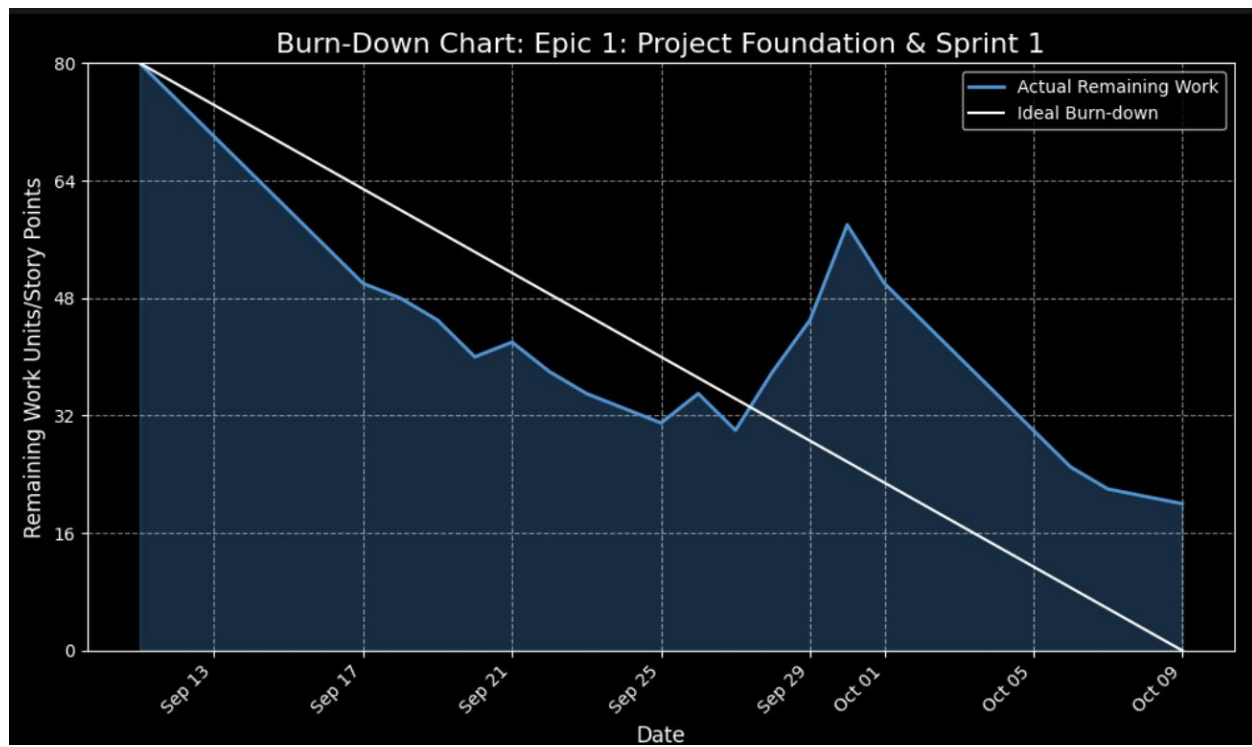


Figure Burndown Chart for sprint 1

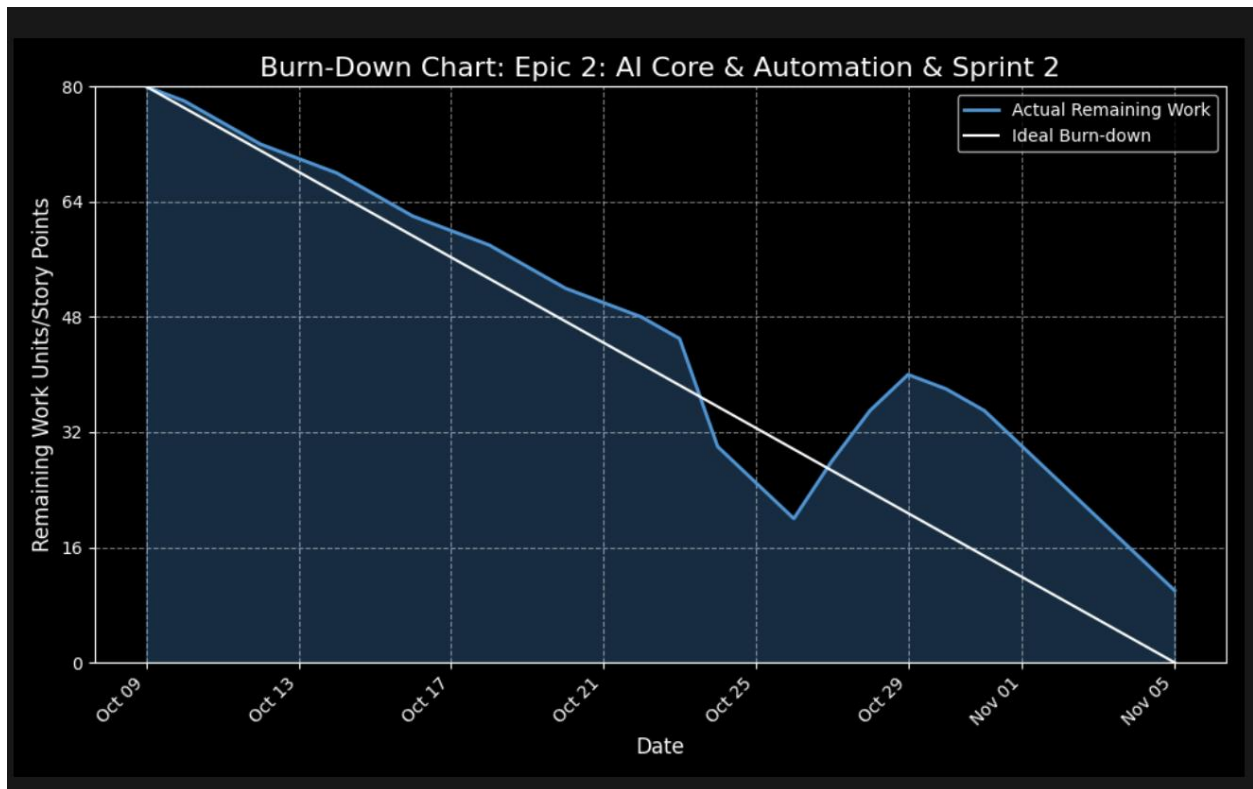


Figure Burndown Chart for sprint 2

The burndown chart for this sprint does not trend towards zero as is typical. This is a direct and expected reflection of the team's strategic decision to pivot from the planned, pointed user stories to the critical, unplanned architectural refactoring. The chart visually confirms this deviation, highlighting the team's adaptability in prioritizing the project's long-term health over short-term feature completion.

5. Sprint Review

Given that this sprint's primary outcome was a critical architectural refactoring rather than new user-facing features, a formal client demonstration was deferred. We kept our supervisor apprised of the necessity and progress of this pivot through regular informal updates.

In retrospect, we recognise that a formal checkpoint meeting with our supervisor to officially ratify such a significant deviation from the sprint plan would have been best practice. This has been a key lesson and we have already incorporated a mandatory weekly supervisor checkpoint into our process to ensure formal alignment on all future progress and pivots. The team continues to operate on the positive feedback from the Sprint 1 review, which validated our project's direction and deemed the initial increment fully deliverable.

6. Retrospect (Critical Review of The Process)

The team demonstrated excellent component ownership and adaptability. Identifying that the original structure was a significant blocker and collectively pivoting to fix it showcases a high level of project maturity. The primary challenges this sprint was the realisation that our initial multi-repository structure was a direct path to realising **Risk R4 - Integration Issues** from our project plan (section 5).

The scattered nature of the codebases was making dependency management difficult and would have led to "significant delays late in the project." The decision to refactor to a monorepo was a direct and powerful mitigation strategy for this specific risk. **In making this pivot, we consciously accepted a new, short-term risk: a temporary reduction in feature-delivery velocity, as reflected in our burndown chart. We deemed this an acceptable trade-off for the long-term health and scalability of the project.** Throughout this technical work, ethical and security considerations remained paramount.

7. Lessons Learned (Critical Review of Sprint One Experience and Future Plan)

This sprint provided several critical lessons in project management and technical strategy:

I. **Architecture Precedes Velocity**

A scalable architecture is a prerequisite for sustained development speed. Investing time to fix a foundational issue is more valuable than building features on a weak base.

II. **Explicit Risk Management**

Proactively identifying that a current process is leading towards a documented risk allows for strategic, justifiable pivots in the sprint plan.

III. **Process Adherence**

Even during technical sprints, a formal communication checkpoint with the project supervisor is crucial for ensuring alignment and documenting major strategic decisions.

IV. **The Importance of Estimating Emergent Work**

We also learned that even large, unplanned technical tasks like the refactor should be estimated in story points after they are identified. Assigning points to this emergent work would have allowed our burndown chart to more accurately reflect the significant effort expended during the sprint, providing a truer measure of our velocity.

These lessons directly inform our future plan. With the architecture now stable, we will prioritise the postponed automation tasks (T4.1 and T4.2) in Sprint 3. Furthermore, two concrete process improvements will be implemented. First, to prevent a recurrence of the credential leak from Sprint 1, we will integrate a *gitleaks* pre-commit hook into our CI/CD pipeline to automatically block any commits containing sensitive keys. Second, a mandatory weekly supervisor checkpoint will be added to our meeting cadence to ensure continuous formal alignment on our progress and plans.